CORE MATHEMATICS GRADE 10: REFLECTION AND CONGRUENCE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.2: Reflection and Congruence
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Symmetry — identifying lines and planes of symmetry in plane figures and objects from the environment – Reflection — properties of reflection in different situations – Equations of Mirror Lines — determining the equation of a mirror line given an object and its image – Congruence — congruence tests for triangles (SSS, SAS, AAS/SAA/ASA, RHS) – Application of Reflection and Congruence — real-life contexts including driving mirrors and road safety
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) identify lines of symmetry in plane figures, b) determine the properties of reflection in different situations, c) draw an image given an object and a mirror line on a plane surface and Cartesian plane, d) determine the equation of the mirror line given an object and its image, e) carry out congruence tests for triangles, f) promote use of reflection and congruence in real-life situations.
Core Competencies	– Learning to Learn: the learner applies the properties of reflection to generate the image of an object on a given mirror line, building independent reasoning skills by moving from physical mirror exploration to abstract Cartesian constructions. – Creativity and Imagination: the learner locates the image of an object placed at a point in front of a mirror and creates paper cutouts of identical shapes to explore direct and opposite congruence.
Core Values	– Responsibility: the learner takes care of the mirror used in reflecting corresponding images, and responsibly collects and observes objects from the immediate environment to identify lines and planes of symmetry. – Respect: the learner recognises the input of every member of the team as they work in groups to make paper cutouts of different identical shapes to identify direct or opposite congruent shapes.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners pose questions about why a reflected image appears the same distance behind a mirror as the object is in front, driving investigation into the properties of reflection. – Planning and Carrying Out Investigations: learners design fold-and-trace investigations using tracing paper and mirror lines to empirically generate the properties of reflection before formalising them algebraically. – Analysing and Interpreting Data: learners measure perpendicular distances, compare side lengths and angles in object-image pairs, and use findings to establish congruence tests (SSS, SAS, AAS, RHS).

	– Constructing Explanations: learners articulate why reflected triangles are congruent and connect the algebraic equation of the mirror line to geometric constructions on the Cartesian plane. – Obtaining, Evaluating and Communicating Information: learners discuss real-life applications (driving mirrors, matatu rear-view mirrors along Thika Road) and communicate findings on road safety to peers.
Pertinent & Contemporary Issues (PCIs)	– Road Safety: the learner discusses with peers the applications of reflections and congruence on images formed by driving mirrors in enhancing road safety — contextualised through matatu and bus mirrors on Kenyan highways such as the Nairobi–Mombasa road. – Environment and Technology: the learner selectively and cautiously collects and observes different objects from the immediate environment (e.g., leaves, Maasai beadwork patterns, door frames in Kisumu market stalls) and identifies the lines and planes of symmetry.
Career Connections	– Architecture and Interior Design: reflection and symmetry underpin building design; Kenyan architects designing structures such as Nairobi's Kenyatta International Convention Centre use symmetry principles in façade planning. – Engineering and Manufacturing: congruence tests are essential in quality control — e.g., ensuring machined parts at the Thika-based manufacturing plants are identical replicas. – Graphic Design and Digital Art: reflection transformations are fundamental tools in design software used by Kenyan creatives producing logos, Maasai-inspired textile prints, and packaging for brands like Brookside. – Transport and Road Safety Planning: understanding mirror images helps matatu and bus drivers interpret driving mirror projections accurately, a competency promoted by the National Transport and Safety Authority (NTSA). – Surveying and Cartography: reflection geometry is applied in land surveying around the Rift Valley for farm boundary demarcation and map-making.
Focus for Lessons	This unit develops learners' understanding of reflection as a geometric transformation — beginning with physical symmetry in everyday Kenyan objects, progressing through the formal properties of reflection on plane surfaces and the Cartesian plane, and culminating in congruence tests for triangles. Learners move from hands-on investigations with plane mirrors and tracing paper to algebraic determination of mirror-line equations, grounding abstract mathematics in observable, real-world phenomena. The unit consistently connects to Kenyan contexts — from the bilateral symmetry of a maize leaf to the safety role of rear-view mirrors on Nairobi's busy roads — ensuring learners appreciate both the beauty and practical utility of reflection and congruence.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a matatu's rear-view mirror show an image that is the same size and shape as the real object behind it — yet appears flipped — and how can we use this to prove that the original and its mirror image are always congruent? KICD KEY INQUIRY QUESTIONS: 1. How do we use reflection in day-to-day life? 2. Where do we use congruence in real life?
Anchoring Phenomenon	A matatu conductor holds up a small rectangular mirror on the Nairobi–Thika highway so the driver can see vehicles approaching from behind. Learners notice that the image of a lorry in the mirror appears to be as far behind the mirror surface as the actual lorry is in front of it, that the image is the same size as the object, and that the image is laterally flipped — the lorry's right headlight appears on the left in the mirror. Yet despite being "flipped," the mirror image of the lorry looks exactly like the lorry in every measurable dimension. This is puzzling: if the image is a perfect copy (same shape, same size, same angles), why does it appear reversed? And if two triangles cut from the same paper template are placed one as the "object" and one as the "mirror image," why can one sometimes be flipped onto the other perfectly (direct congruence) while other times it cannot without lifting it off the page (opposite/indirect congruence)? These observations anchor the entire unit: learners will investigate the properties that make reflected images predictable, determine mirror-line equations, and establish the minimum conditions (congruence tests) needed to guarantee two triangles are identical in shape and size.

Storyline Thread	Lesson 1 — Symmetry All Around Us (Anchoring Phenomenon + Lines of Symmetry): Learners observe the anchoring phenomenon (matatu mirror image) and supporting phenomena (maize leaf, Lake Victoria reflection). Working in pairs, they collect objects from the school environment (leaves, bottle caps, exercise book covers) and use a plane mirror to locate lines of symmetry. They record findings on a class Driving Question Board (DQB): "What makes the two halves match perfectly?" Formally, learners identify lines of symmetry in plane figures (equilateral triangle, rectangle, rhombus, regular hexagon) and distinguish between figures with one, many, or no lines of symmetry. Kenyan context: symmetry patterns in Maasai beadwork and kikoi fabric designs. Lesson 2 — Properties of Reflection (Physical Investigation): Learners use a plane mirror, tracing paper, and a drawn object (an arrow representing a matatu) to discover the four properties of reflection: (i) the image is the same size as the object; (ii) the mirror line is the perpendicular bisector of the line joining each point to its image; (iii) the image is laterally inverted; (iv) object and image are equidistant from the mirror line. Groups fold tracing paper along the mirror line to verify coincidence. Learners update the DQB: "The mirror line is always in between — and at equal distances!" Connecting to the driving question: this explains why the lorry in the matatu mirror appears as far behind the glass as it truly is in front. Lesson 3 — Drawing Images on a Plane Surface (Applying Properties): Learners draw objects (a flag outline representing Kenya's flag shape) and given mirror lines on plain paper. Using the perpendicular bisector property established in Lesson 2, they construct the reflected image step by step — marking perpendiculars from key vertices, measuring equal distances, and connecting image points. Teacher emphasises accuracy with ruler and set square. Learners reflect irregular polygons and record: "Every point reflects to a unique image point across the mirror line." DQB updated with a student-drawn diagram showing object–mirror line–image. Lesson 4 — Reflection on the Cartesian Plane (Mirror Lines $x = k$, $y = k$, $y = x$, $y = -x$): Learners draw objects on a Cartesian grid and reflect them in horizontal ($y = 0$, $y = k$), vertical ($x = 0$, $x = k$), and diagonal ($y = x$, $y = -x$) mirror lines. Working in groups, they tabulate coordinate pairs of object and image vertices and identify the algebraic rule for each mirror line type. Connecting to the anchoring phenomenon: "The matatu mirror is like the y-axis — the image appears directly opposite." Learners practise at least three examples on squared paper and annotate coordinate transformations. DQB updated: "We can describe where the image lands using coordinates and algebra." Lesson 5 — Finding the Equation of the Mirror Line: Given an object and its image on the Cartesian plane, learners reverse-engineer the mirror line. They construct the perpendicular bisector of the segment joining corresponding vertices to locate the mirror line, then determine its equation (gradient and y-intercept). Problem contexts are Kenyan: a footpath reflected in a river (representing the Tana River valley), a road sign reflected in a puddle along Mombasa Road. Learners discover that the mirror line is always the perpendicular bisector locus and use the midpoint and gradient formulae to write its equation. DQB updated: "If we know the object and image, we can work backwards to find the mirror." Lesson 6 — Introduction to Congruence and Direct vs. Opposite Congruence: Learners make paper cutouts of triangles and quadrilaterals (connecting to chapati-cutter supporting phenomenon). They explore whether cutouts can be made to coincide by sliding and rotating only (direct/positive congruence) or whether flipping is needed (opposite/indirect congruence — the situation with the matatu mirror image). Learners classify pairs of shapes and record observations. The teacher introduces the formal definition of congruence: same shape, same size, all corresponding sides and angles equal. DQB updated: "Reflected images are always congruent to their objects — but they might be mirror images of each

	other!" Lesson 7 — Congruence Tests for Triangles (SSS, SAS, AAS, RHS): Learners use different triangles (drawn on plain paper) to investigate the minimum conditions needed to guarantee two triangles are congruent. Working in groups, they attempt to construct non-congruent triangles given: (i) three sides — SSS; (ii) two sides and included angle — SAS; (iii) two angles and a side — AAS/SAA/ASA; (iv) right angle, hypotenuse and side — RHS. Groups present findings: each test yields exactly one triangle (up to congruence). Kenyan context: a carpenter in Gikomba market cutting identical triangular roof trusses — how many measurements are the minimum needed to guarantee all trusses are identical? DQB updated: "We only need THREE matching parts — but they must be the right three!" Lesson 8 — Applications, Consolidation, and Model Revision: Learners revisit the anchoring phenomenon fully: they now explain (using properties of reflection) why the lorry's mirror image is congruent to the actual lorry, why it appears laterally inverted, and which congruence test (SAS, using the mirror line as the axis of symmetry) proves the triangles formed by object–mirror line–image are congruent. Learners complete a group task: analysing driving mirror geometry (boda boda and matatu contexts) to justify NTSA road-safety guidance on mirror placement. A cumulative Cartesian model is produced — showing object, mirror line, image, perpendicular bisectors, labelled equations, and congruence annotations. DQB is reviewed: all initial questions are answered and remaining wonderings are noted for future strands. Formative assessment via class activity and a short written test.
Supporting Phenomena	– Symmetry in a maize leaf (Zea mays): learners in Kericho or Nakuru observe that a maize leaf can be folded along its central vein so that the two halves match perfectly — a physical demonstration of a line of symmetry and the concept that reflection maps every point on one side to a corresponding point equidistant on the other. – Reflection in Lake Victoria: photographs taken at sunrise near Kisumu show the hills of Homa Bay perfectly reflected on the still water surface, prompting learners to identify the waterline as the mirror line and measure that the reflected hills appear at the same apparent distance below the surface as the real hills rise above it. – Identical paper-cutout chapati shapes: a cook in a Nairobi restaurant cuts chapati dough using a circular cutter — every cutout is congruent to every other. Learners explore whether rotating, flipping, or sliding one cutout can always make it coincide with another, introducing direct vs. opposite congruence in a familiar food context. – Rear-view and side mirrors on a bicycle (boda boda): a boda boda rider in Kisumu adjusts the handlebar mirror; learners observe that the mirror line (the mirror surface) is perpendicular to the line joining the object (following vehicle) and its image, and that the image distance equals the object distance — directly establishing the perpendicular bisector property of reflection.

LESSON 1 (40 minutes): Symmetry All Around Us — Anchoring the Matatu Mirror Phenomenon & Lines of Symmetry	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor learners in the matatu rear-view mirror phenomenon and build foundational understanding of lines of symmetry in plane figures as the entry point into reflection and congruence.
Knowledge	– Define a line of symmetry as a line that divides a plane figure into two mirror-image halves. – Identify and count lines of symmetry in regular and irregular plane figures (triangles, rectangles, squares, circles, letters of the alphabet). – Distinguish between figures that have symmetry and those that do not, using the matatu mirror context as anchor.
Skills	– Fold and trace paper cut-outs to locate lines of symmetry in plane figures. – Record observations systematically in a symmetry table and begin to articulate what 'same shape, same size, flipped' means geometrically.
Attitudes	– Appreciate that mathematical symmetry is embedded in everyday Kenyan objects — matatu art, Maasai beadwork patterns, the Kenya flag, architectural designs. – Show responsibility in handling mirrors and paper cut-outs as learning resources.
Key Inquiry Question	How do we use reflection in day-to-day life?
Purpose in Storyline	This is the opening lesson of the unit. It introduces the anchoring phenomenon — the matatu conductor's rear-view mirror on the Nairobi–Thika highway — and uses it to surface learners' intuitive ideas about symmetry, flipping, and sameness. By the end of the lesson learners have opened the Driving Question Board with their own puzzling questions and have drawn a first rough model of what a mirror does to a shape. Every subsequent lesson will return to this board and this model.
Safety Notes	Handle small plane mirrors carefully — edges can be sharp. Do not direct mirror reflections into other learners' eyes. If using printed images, scissors should be handled with care during paper-folding activities.

B. LESSON OVERVIEW

This lesson launches the entire sub-strand by presenting learners with a vivid, locally grounded phenomenon: a matatu conductor on the Nairobi–Thika highway holds a small rectangular mirror so the driver can see a lorry approaching from behind. Learners are immediately confronted with three puzzling observations — the image appears as far behind the mirror as the lorry is in front of it, the image is the same size as the real lorry, and yet the lorry's right headlight appears on the left in the mirror. The lesson does not resolve these puzzles; instead, it uses them to generate genuine mathematical curiosity and to open the Driving Question Board.

The bulk of the lesson is spent on a hands-on symmetry exploration. Working in pairs, learners receive paper cut-outs of common plane figures — an equilateral triangle, a rectangle, a square, the letter "A", the letter "F", and a rough outline of a matatu — and fold them to discover lines of symmetry. This low-floor, high-ceiling activity connects directly to the phenomenon: a line of symmetry is precisely a mirror line, and the two halves of a symmetric figure are the first example of a reflected image being congruent to its object. Learners record findings in a class symmetry table on the board.

The lesson closes with learners writing their first wonder questions on sticky notes for the Driving Question Board and sketching a rough initial model (a diagram with labels) that captures what they currently think a mirror does to a shape. This model will be revised in every subsequent lesson as understanding deepens. Throughout,

the teacher uses the matatu context, references to Maasai beadwork, the Kenyan flag, and matatu body-art patterns to keep the mathematics grounded in Kenyan daily life.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Number Lines: Football Game (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a large printed image (or teacher sketch on the board) of a matatu on the Nairobi–Thika highway with a conductor holding a small rectangular mirror so the driver can see an approaching lorry. Each learner writes three individual predictions in their exercise books: (1) What does the lorry's image in the mirror look like compared to the real lorry? (2) If you measured the height of the image and the height of the real lorry, what would you find? (3) Why might the lorry's right headlight appear on the LEFT side in the mirror? After 3 minutes of individual thinking, learners share predictions with a partner (Think-Pair) before selected pairs share with the class.	Display the matatu mirror image/sketch and say: 'Angalieni picha hii kwa makini — look at this picture carefully. A conductor on a matatu travelling from Nairobi towards Thika holds this mirror. The driver can now see a lorry behind the matatu. Before I tell you ANYTHING about reflection or symmetry, I want YOU to tell ME what you think is happening.' Pause — allow 10 seconds of silent looking. Then say: 'In your exercise book, write THREE predictions. Number them 1, 2, 3. You have 3 minutes. Go.' [Allow 3 minutes of silent individual writing. Circulate — do NOT correct, only observe the range of ideas.] After 3 minutes: 'Turn to your partner and compare your predictions. Do you agree? Where do you disagree? One minute.' [1 minute pair talk.] Cold-call 4 pairs: select one pair who predicted the image is the same size, one who predicted it would be smaller, one who noticed the flip, and one who had a different idea. For each, ask: 'What is your reasoning — kwa nini unafikiria hivyo?' [Wait 10–12 seconds after each response before affirming or redirecting.] Record key prediction phrases verbatim on the board under the heading 'Tunafikiria / We think…' Do NOT evaluate predictions as right or wrong.	Think-Pair-Share with Prediction Recording. This strategy makes prior knowledge visible and creates cognitive dissonance — learners discover that their peers hold contradictory ideas about mirror images (same size vs. smaller, flipped vs. not flipped). The recorded predictions become a public accountability tool: learners will return to them at the end of the unit to see how their thinking has changed.	Teacher circulates and looks for: (a) whether learners are thinking about SIZE (same/smaller/larger), ORIENTATION (flipped/not flipped), and DISTANCE (how far behind the mirror the image appears) — these three dimensions signal readiness to explore reflection properties. (b) Learners who write 'the image is smaller' — note these learners; they may be confusing reflection with reduction/enlargement (Sub-strand 2.1). (c) Learners who notice the lateral flip without prompting — these are strong entry points for the DQB question about congruence despite flipping. Flag at least 2 written predictions to read aloud during Explain phase.

Observe Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Number Lines: Football Game (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each pair receives an envelope containing six paper cut-outs: (1) an equilateral triangle, (2) a rectangle (non-square), (3) a square, (4) the capital letter 'A', (5) the capital letter 'F', and (6) a rough rectangle labelled 'matatu' with a small rectangle on one end labelled 'mirror'. Learners fold each shape to find lines of symmetry, mark each fold line with a pencil, and record results in a class-format table drawn in their exercise books with columns: Shape | Number of lines of symmetry | Sketch of the line(s). Learners have 8 minutes for this hands-on activity. Those who finish early are challenged: 'Can you find the line(s) of symmetry in a regular hexagon? Draw one and investigate.'	Distribute envelopes. Say: 'Each envelope has six shapes. Your job: fold each shape so that the two halves match PERFECTLY — one half lands exactly on top of the other. Each fold line you find is called a LINE OF SYMMETRY. Mark it with your pencil. Record your findings in the table in your book.' [Demonstrate ONE fold of the square so learners understand the task — do not show all answers.] Set a visible 8-minute timer. Circulate actively. When you see a pair struggling with the letter 'F': crouch to their level and ask 'When you fold it this way, do the two halves match?' — wait 12 seconds for them to check before saying anything further. When you see a pair who has found that 'F' has NO lines of symmetry, ask 'Interesting — why do you think that is? What is different about F compared to A?' [Wait 10 seconds.] At 6 minutes, give a 2-minute warning. At 8 minutes, call the class to attention. Cold-call 5 different pairs to report one shape each — equilateral triangle, rectangle, square, letter A, letter F — asking each: 'How many lines did you find, and where are they?' Build the class results table on the board. Highlight the 'matatu' cut-out last: 'Now look at the matatu shape. Does it have a line of symmetry? What does that tell us about the mirror the conductor is holding?'	Concrete Manipulative Exploration with Structured Data Collection. Folding physical paper cut-outs gives learners kinesthetic, visual, and tactile evidence of symmetry — far more convincing than a diagram in a textbook. The structured table forces learners to translate physical action into mathematical notation (number, sketch). Connecting the 'matatu' cut-out back to the phenomenon at the end of the activity anchors the abstract concept in the opening story.	Teacher looks for: (a) Correct fold lines — equilateral triangle (3 lines), rectangle (2 lines), square (4 lines), letter A (1 line), letter F (0 lines). Any pair reporting 4 lines for a rectangle needs immediate redirection — ask them to fold along a diagonal and check whether the halves match. (b) Quality of sketches — do learners draw the line of symmetry accurately through the figure, or do they place it outside the shape? (c) Pairs who correctly identify that 'F' has no lines of symmetry and can explain WHY — these learners are ready for the deeper question about 'opposite congruence' later in the unit. (d) Whether ANY learner spontaneously connects the fold line to 'the mirror line' — if so, amplify this publicly.
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12	Learners return to the completed class table on the board. The teacher leads a whole-class discussion that builds a shared definition of 'line of symmetry' and explicitly connects it to the matatu mirror phenomenon. Learners then	Point to the class table and say: 'Let us make sense of what we just found. Look at the square — 4 lines. Look at the letter F — zero lines. What is a LINE OF SYMMETRY? I want someone to try a definition in their own words	Socratic Discussion into Formal Definition (Concept Attainment). Rather than presenting the definition first, the teacher draws the definition OUT of learners' own words from the observation activity, then formalises it. This	Teacher listens for: (a) Whether learners' verbal definitions include both 'divides into two halves' AND 'mirror image / congruent' — a definition that only says 'divides into two equal halves' is incomplete (a diagonal of a non-

🔍 Search ARES for similar videos 📖 **HTML:** Number Lines: Football Game (PLIX) Source: CK-12 🔍 Search ARES for similar readings	copy a formal definition and a worked example into their notes. In the final 3 minutes of this phase, learners individually attempt one consolidation task: given a drawing of a Kenyan flag (which has a vertical axis of symmetry through the Maasai shield), identify and draw all lines of symmetry and state whether the flag has any.	before I give you the formal one.' Cold-call 3 learners. [Wait 12 seconds after each.] After the third response, synthesise: 'A line of symmetry is a line that divides a plane figure into two congruent halves — each half is a mirror image of the other. Write that down.' Write the definition on the board. Then point back to the matatu image: 'Now — the conductor's mirror. If I stand a mirror upright on the line of symmetry of this rectangle, what do I see? I see the LEFT half REFLECTED to look exactly like the RIGHT half. The mirror IS the line of symmetry.' Pause 15 seconds. Ask: 'So when the mirror shows the lorry behind the matatu — what is the mirror acting as?' [Wait 10 seconds — cold-call 2 learners.] Expected response: 'a line of symmetry' or 'the mirror line.' Affirm and add: 'Exactly. And later in this unit we will call it the MIRROR LINE — the line of reflection.' Now say: 'Open to a fresh page. I will show you the Kenyan flag. Find its lines of symmetry and sketch them.' [Draw/display Kenyan flag outline — 3 horizontal bands, Maasai shield and spears in the centre.] Allow 3 minutes individual work. Cold-call one learner to come to the board and draw the line(s). Discuss: 'Does the flag have a horizontal line of symmetry? Check carefully — are the top and bottom halves mirror images?' [The flag has ONE vertical line of symmetry through the centre of the shield; the horizontal bands are different colours so there is no horizontal symmetry — this is a productive discussion point.]	'concept attainment' approach means learners own the idea before it is formalised. The Kenya flag example immediately applies the definition in a new, culturally relevant context, testing transfer.	square rectangle divides it into two congruent triangles but is NOT a line of symmetry). Probe with: 'I draw a line from corner to corner of this rectangle — does that divide it into two equal halves? Is it a line of symmetry? Why not?' (b) On the Kenya flag task: any learner who correctly identifies that the horizontal axis is NOT a line of symmetry (because red ≠ green) is demonstrating careful analytical thinking — praise this publicly and connect it to 'we need BOTH shape AND orientation to match, which is exactly what we will explore with congruence.'

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Properties of Equality and Congruence Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Number Lines: Football Game (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or tears a small piece of paper into two). On the FIRST sticky note they write one question that the matatu mirror phenomenon makes them wonder about — something they genuinely do not yet know. On the SECOND sticky note they write one thing they noticed (an observation) from today's activity that they cannot yet fully explain. Learners bring their sticky notes to a designated space on the classroom wall (or a large sheet of manila paper) labelled 'DRIVING QUESTION BOARD — Why does the matatu mirror show a flipped but identical image?' The teacher organises the sticky notes into emerging clusters in front of the class.	Say: 'Today we have seen something interesting: a matatu mirror shows an image that is the same size as the real object but appears flipped. We found lines of symmetry in flat shapes. But we have MORE questions than answers right now — and that is exactly where good mathematics begins.' Distribute sticky notes. Say: 'On your PINK note — or first piece of paper — write ONE question. It must be something you genuinely wonder about and cannot yet answer. It might be about the mirror, about the flip, about size, about the letter F having no symmetry — anything connected to what we did today. On your BLUE note — or second piece — write ONE observation you cannot yet explain. You have 3 minutes. Write in full sentences.' [Allow 3 minutes of silent writing. Circulate and read over shoulders — gently prompt any learner who is stuck: 'What surprised you today?' or 'What would you like to know more about?'] After 3 minutes, learners bring notes to the board. As they post them, teacher reads 6–8 aloud and begins clustering: group questions about SIZE together, questions about the FLIP together, questions about DISTANCE together, questions about CONGRUENCE/SAMENESS together. Say: 'This board is OURS for the whole unit. Every lesson, we will add to it or move things around as our understanding grows. By the final lesson, we should be able to answer every single question on this board.' Leave the board visible for the rest of the unit. Point to the driving question written at the top	Driving Question Board (DQB) Launch. The DQB is a physical, persistent record of the class's collective curiosity. Organising questions into clusters (size, flip, distance, congruence) previews the conceptual structure of the unit without foreclosing inquiry. Learners see that their personal questions matter and are part of a shared investigation — this builds ownership and motivation. The board also serves as a formative assessment tool across the unit: the teacher can track which clusters get answered first and which persist as productive tensions.	Teacher reads all sticky notes (before or after class). Look for: (a) Questions that reveal misconceptions — e.g. 'Does the mirror make the lorry smaller?' (confusing reflection with enlargement) or 'Is the image behind the mirror real?' (ray optics confusion) — note these for targeted addressing in later lessons. (b) Questions that anticipate upcoming content — e.g. 'How do you find the equation of the mirror line?' or 'Can you always flip a shape onto its mirror image?' — these learners can be given extension challenges in later lessons. (c) Any learner who has NOT written a genuine question (wrote 'I don't know' or left it blank) — follow up individually: 'Tell me one thing that surprised you today' to draw out a verbal question, then help them write it.

		of the board: 'Why does the matatu mirror show an image that is the same size and shape yet appears flipped — and how can we prove the original and its mirror image are always congruent?' Say: 'This is our BIG question. Keep it in your mind.'		
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Number Lines: Football Game (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners draw a first 'Initial Model' in their exercise books. The prompt is displayed on the board: 'Draw a diagram that shows what you think a mirror does to a shape. Label: the OBJECT (the real thing), the MIRROR LINE, and the IMAGE (what appears in the mirror). Use the lorry-and-mirror situation as your example. Add at least two labels that describe what you notice about the relationship between the object and the image.' This is an individual task (5 minutes). Learners should draw freely — this model does NOT need to be correct. It is a record of current thinking.	Say: 'Before we end today, I want each of you to draw what you currently think is happening when an object is placed in front of a mirror. This is your INITIAL MODEL — think of it as a scientific sketch that you will improve over the next several lessons. It does not need to be perfect. It needs to be HONEST — draw what YOU think, not what you think I want to see.' Write the prompt on the board. Say: 'You have 5 minutes. Label at least: the object, the mirror line, and the image. Add two observations as written labels anywhere on your diagram.' [Allow 5 minutes of silent individual drawing. Circulate — observe but do not correct. Encourage with: 'Keep going — what else do you notice about the sizes?' or 'Can you add a label about the distance?'] At 4 minutes, say: 'One more minute — add at least one question mark anywhere on your model for something you are still unsure about.' At 5 minutes: 'At the top of your model, write today's date and the words: INITIAL MODEL — Lesson 1. We will look at these again at the end of the unit and you will see how much your thinking has grown.' Cold-call 2 volunteers to hold up and briefly describe their model (30 seconds each). Do NOT evaluate as right or wrong — respond with: 'Thank you — I can see you are thinking about	Initial Model Construction (Modelling as a Tool for Thinking). Drawing a model at the START of a unit, before instruction is complete, serves a different purpose from a model drawn at the end. It externalises current mental models, making implicit ideas visible and available for revision. The question-mark instruction deliberately signals that uncertainty is mathematically productive. Labelling the model 'Initial Model — Lesson 1' sets up the cumulative model-revision practice that will run through the entire unit.	Collect (or photograph) models before the next lesson. Assess for: (a) Whether learners draw the image on the OPPOSITE side of the mirror line from the object — this is the key geometric property. Any learner who draws the image on the SAME side has a fundamental misconception to address in Lesson 2. (b) Whether learners show the image as the same size as the object, or as different — size conservation of reflection is a key SLO. (c) Quality of labels — do learners use words like 'flipped,' 'same size,' 'equal distance'? These indicate readiness for formal vocabulary. (d) Question marks — what residual uncertainties do learners flag? Use these to inform the sequence and pacing of Lessons 2–6.

		[size/distance/flip]. We will investigate that in our next lesson.'		

D. TEACHER REFLECTION
1. During the Predict Phase, how wide was the range of predictions? Did most learners focus on SIZE, or did they also notice the FLIP and DISTANCE aspects of the phenomenon? What does this tell me about which properties of reflection I need to spend more time on in Lesson 2?
2. In the Observe Phase, were learners able to correctly identify that the letter 'F' has NO lines of symmetry? Did any pair struggle to distinguish between 'divides into equal halves' and 'creates a mirror image'? How will I address this distinction explicitly in the next lesson?
3. When I connected the fold line to the mirror line during the Explain Phase, how many learners made the connection spontaneously before I pointed it out? What does this indicate about their intuitive understanding of reflection as a transformation?
4. Looking at the sticky notes on the Driving Question Board: which cluster has the most questions — size, flip, distance, or congruence? Does the distribution of questions match the conceptual sequence I have planned for Lessons 2–6, or do I need to reorder lessons to address the most pressing learner questions first?
5. Reviewing the Initial Models collected at the end of the lesson: what proportion of learners drew the image on the correct (opposite) side of the mirror line? For those who did not, is this a drawing error or a genuine conceptual misconception — and what is my plan to address it at the opening of Lesson 2?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that a matatu conductor's mirror on the Nairobi–Thika highway produces an image of a lorry that appears the same size as the real lorry yet is laterally flipped (the lorry's right headlight appears on the left in the mirror). Through paper-folding, learners also observed that plane figures can have different numbers of lines of symmetry — a square has 4, a rectangle has 2, an equilateral triangle has 3, the letter 'A' has 1, and the letter 'F' has 0.
What did I learn?	A line of symmetry is a line that divides a plane figure into two congruent halves, each of which is a mirror image of the other. The mirror held by the matatu conductor acts as a mirror line (line of reflection). Figures with lines of symmetry include most regular polygons and many Kenyan cultural objects (Maasai beadwork, the Kenya flag), while asymmetric figures like the letter 'F' have none. The fold line in a paper cut-out is equivalent to the mirror line in a reflection.
How does this explain the phenomenon?	The fact that the matatu mirror produces an image that is the same size as the real lorry — even though it appears flipped — is the first clue that reflection preserves shape and size (congruence). When we folded a rectangle along its line of symmetry, the two halves matched perfectly: same shape, same size, but one was a 'flip' of the other. This is exactly what the mirror does to the lorry. The puzzle of why the flipped image is still the same size and shape as the original will be the central question we investigate throughout this unit.

LESSON 2 (40 minutes): Reflection All Around Us — Locating Lines of Symmetry in Plane Figures	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use physical mirrors and paper-folding to locate and count lines of symmetry in objects collected from the school environment and in standard plane figures, connecting the idea of 'perfect halves' to the matatu mirror phenomenon.
Knowledge	– Define a line of symmetry as a line that divides a plane figure into two congruent, mirror-image halves. – Identify and count the exact number of lines of symmetry in an equilateral triangle, rectangle, rhombus, square, and regular hexagon. – Distinguish between figures that have one line of symmetry, many lines of symmetry, or no line of symmetry.
Skills	– Use a plane mirror and paper-folding to locate lines of symmetry in real objects (leaves, bottle caps, exercise book covers) and drawn figures. – Record and compare findings systematically in a table, using correct geometric language.
Attitudes	– Appreciate that symmetry is a mathematical idea embedded in Kenyan cultural artefacts such as Maasai beadwork and kikoi fabric designs. – Show responsibility by handling the mirror carefully and respecting peers' contributions during group work.
Key Inquiry Question	How do we use reflection in day-to-day life?
Purpose in Storyline	Lesson 1 established that the matatu mirror image looks exactly like the lorry — same shape, same size. Lesson 2 deepens this by asking WHY the two halves match perfectly: learners discover that a line of symmetry is precisely the mirror line, and that folding along it maps one half exactly onto the other — the first step toward understanding that reflected images are always congruent.
Safety Notes	Handle small plane mirrors with care — edges can be sharp. Learners should place mirrors flat on the desk rather than holding them at face level. No glass mirrors should be broken; if a mirror cracks, the teacher disposes of it immediately. Collected leaves should be from non-toxic plants only; wash hands after handling.

B. LESSON OVERVIEW

This lesson places learners in the role of mathematical investigators, using the anchoring phenomenon — the matatu rear-view mirror on the Nairobi–Thika highway — as the motivating puzzle. In Lesson 1, learners established that the mirror image of the lorry appears identical in size and shape to the real lorry. Lesson 2 now asks: what is the special line that makes the two halves match so perfectly? Working in pairs, learners collect objects from the school environment (maize leaves, bottle caps, exercise book covers) and use a small plane mirror to search for lines of symmetry, physically observing that placing the mirror along a line of symmetry produces an image that completes the original figure seamlessly — exactly what the matatu conductor's mirror does for the driver on the highway.

In the Observe and Explain phases, learners move from informal object exploration to formal geometric figures. They fold cut-out shapes — equilateral triangle, rectangle, rhombus, and regular hexagon — to find and draw every line of symmetry, then record results in a class table. The Kenyan cultural context is woven in: learners examine photographs of Maasai beadwork collars and kikoi fabric strips and identify how the symmetry patterns are constructed, connecting abstract mathematics to the textiles of their own communities.

By the end of the lesson, learners can state, in their own words, that a line of symmetry is a mirror line — the same kind of line that separates the lorry from its image in the matatu mirror. This precise language anchors the Driving Question Board update and advances the unit's central argument: that reflection along a mirror line

preserves shape and size, which is the foundation of congruence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Number Lines: Football Game (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a printed or hand-drawn outline of a maize leaf and is asked, WITHOUT folding or using a mirror: 'Draw a dotted line on this leaf where you think you could fold it so that both halves match perfectly.' Learners write one sentence below their drawing predicting how many such lines exist on the leaf. They then do the same for a bottle-cap circle outline. Predictions are kept face-down until the Observe phase to prevent anchoring on peers' ideas.	Distribute the maize leaf and circle outlines. Say: 'Before we touch any mirrors today, I want you to think like the matatu driver — he is predicting where vehicles are based on what he sees in the mirror. You are going to predict where the perfect fold line is on these shapes. Work silently for two minutes.' [WAIT TIME: 2 minutes silent individual work.] After time, ask: 'How many lines does the maize leaf have? Hold up one finger for one line, two fingers for two lines, more fingers for more.' Cold-call 3 learners to justify their finger count: 'Akinyi — you said one line. Where exactly is it?' [WAIT TIME: 10 seconds after each cold-call.] Do the same for the circle: cold-call 3 different learners. Record the class distribution of predictions (0, 1, 2, many) on the board as a tally — do NOT confirm or deny answers yet. Close by saying: 'Keep your predictions — we will check them with mirrors in just a moment.'	Prediction Commitment with Evidence Anchoring — by committing predictions to paper before observing, learners activate prior intuitive knowledge about symmetry and create a personal record to compare against evidence, making the subsequent observation more cognitively impactful.	Teacher scans held-up fingers to gauge prior knowledge spread. Listen for whether learners mention 'fold,' 'mirror,' or 'equal halves' unprompted — this indicates readiness to connect to formal vocabulary. Flag learners who predict zero lines for the circle (a common misconception) for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Working in pairs, learners receive: one small plane mirror, one maize leaf (real or cut-out), one bottle cap, one exercise book cover (rectangular), and a set of four pre-cut paper shapes (equilateral triangle, rectangle, rhombus, regular hexagon). Activity 1 (5 min): Learners place the mirror on the maize leaf, sliding it around until the reflection in the mirror	Before releasing materials say: 'Remember — the matatu conductor holds the mirror so the driver sees a perfect image of the lorry. You are going to find the position where the mirror makes a perfect image that completes each shape. That position IS the line of symmetry.' Circulate during Activity 1. At the 3-minute mark, pause the class: 'Hands up if your mirror	Physical Manipulation with Systematic Recording — using a real mirror and paper-folding grounds the abstract definition of a line of symmetry in kinesthetic experience, while the structured table forces learners to generalise across multiple cases rather than treating each shape as an isolated observation.	Check tables for completeness and accuracy: equilateral triangle = 3 lines, rectangle = 2 lines, rhombus = 2 lines (along diagonals — note: this is a common error; the rhombus's lines of symmetry ARE its diagonals, NOT lines joining midpoints of opposite sides), regular hexagon = 6 lines. Identify any pair recording the wrong number for the rhombus or

[Number Lines: Football Game (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	completes the leaf perfectly. They draw the mirror position as a dotted line and compare to their prediction. Activity 2 (8 min): Learners fold each paper shape and draw every fold line that creates two perfectly matching halves. They record results in a table with columns: Shape | Number of Lines of Symmetry | Sketch of Lines. Activity 3 (2 min): Learners examine a projected photograph of a Maasai beadwork collar and a kikoi fabric strip and count symmetry lines they can spot, recording in the same table.	position for the maize leaf matched your prediction exactly.' [WAIT TIME: 10 seconds.] Cold-call 1 learner whose hand is up and 1 whose hand is down: 'Kamau — yours matched. Why do you think it matched?' 'Wanjiku — yours didn't match. What surprised you?' [WAIT TIME: 12 seconds each.] During Activity 2, watch for learners trying to fold a rhombus along its diagonals — prompt: 'When you fold along THAT line, do the two halves land exactly on top of each other? Check carefully.' For the hexagon, prompt: 'Try all the fold lines you can find — there may be more than you expect.' During Activity 3, project the Maasai beadwork image for 90 seconds, then the kikoi for 90 seconds. Ask: 'How many lines of symmetry did you find in the kikoi strip?' Cold-call 4 pairs to report, recording their answers on the board in a quick tally.		hexagon for re-engagement in the Explain phase. Also note whether learners describe the mirror position using 'perpendicular' or 'midpoint' language — this previews the formal properties of reflection needed in Lesson 3.
Explain Phase 📹 **VIDEO:** [Genetics vocabulary] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Number Lines: Football Game (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Pairs join into groups of four to compare tables and resolve any disagreements. Each group appoints a 'reporter' to present one disputed case to the class. The class then co-constructs a formal definition: learners suggest words, the teacher refines them on the board. Learners copy the agreed definition and complete two guided tasks in their exercise books: (1) Draw an isosceles triangle and mark ALL its lines of symmetry; (2) Draw a regular pentagon and predict how many lines of symmetry it has before folding a cut-out to check. Learners also write one sentence connecting today's activity to the matatu mirror: 'The mirror line in the	Open whole-class discussion: 'Which shape caused the most disagreement in your group?' Cold-call 3 reporters. [WAIT TIME: 10 seconds after each.] For any disagreement about the rhombus, hold up the cut-out rhombus and fold it live in front of the class along both diagonals, then along the lines joining midpoints of opposite sides — ask the class: 'Which fold gives two halves that land exactly on top of each other? Tell your partner.' [WAIT TIME: 15 seconds pair-talk.] Cold-call 2 learners. Once consensus is reached, write on the board: 'A LINE OF SYMMETRY is a line that divides a plane figure into two congruent halves such that one	Claim–Evidence–Reasoning (CER) Whole-Class Discussion — learners are asked to make a claim (this is/is not a line of symmetry), support it with physical evidence (the fold test), and explain their reasoning in words, building the habit of mathematical justification that underpins congruence proofs later in the unit.	Read 4–5 'matatu connection' sentences as learners write. Look for: (a) correct use of 'mirror line,' (b) mention of 'same size/shape' or 'congruent,' (c) explicit link to the lorry image being a perfect copy. Learners who write vague sentences ('the mirror shows the car') need a follow-up prompt: 'Which PART of the mirror situation corresponds to a line of symmetry?' Flag these learners for the DQB phase.

	matatu mirror is like the line of symmetry because ___.'	half is the exact mirror image of the other.' Say: 'Write this in your exercise book. Underline the word CONGRUENT — we will use it a lot this term.' For the matatu connection sentence, cold-call 3 learners to read theirs aloud. [WAIT TIME: 12 seconds per learner.] Affirm correct connections and gently correct any that conflate the mirror line with the mirror surface itself: 'The LINE on the mirror, not the mirror itself, is what we call the line of symmetry — the mirror is just our tool for finding it.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Genetics vocabulary] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Number Lines: Football Game (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each pair receives two sticky notes (or small paper slips). On the FIRST (green/light-coloured), they write ONE thing they now understand about why the matatu mirror image looks like a perfect copy of the lorry — starting with 'I now know that…'. On the SECOND (yellow/different colour), they write ONE new question the lesson has raised for them — starting with 'I still wonder…' or 'This makes me wonder…'. Pairs post both notes on the class Driving Question Board, grouping them under the teacher-prepared headings: 'What makes the two halves match?' | 'What is the mirror line?' | 'New questions for later lessons.'	Point to the DQB and say: 'In Lesson 1 we asked: why does the lorry's image in the matatu mirror look exactly like the lorry? Today we have a better answer. Let's update our board.' Distribute the two sticky notes. Give 3 minutes for writing. [WAIT TIME: 3 minutes.] As pairs post notes, read 4 'I now know' notes aloud and 3 'I still wonder' notes aloud — choose ones that represent a spread of understanding. For a strong 'I now know' note, say: 'This pair says the mirror line is the line of symmetry of the whole situation — the lorry plus its image. That is a powerful idea we will prove in Lesson 3.' For a rich wonder note — e.g., 'I wonder if every shape has at least one line of symmetry' — say: 'That is a brilliant mathematical question. Let's pin it here — we may answer it before this unit is done.' Do NOT answer all wonder questions immediately; explicitly park them: 'This board is growing. By Lesson 8, we will have answered most of these together.'	Driving Question Board (DQB) Progressive Reveal — the two-colour sticky-note protocol separates consolidated understanding from open questions, making the growth of knowledge visible over the lesson sequence and sustaining curiosity as an epistemic driver rather than letting unanswered questions become sources of confusion.	Read all posted 'I now know' notes during or immediately after the lesson. Look for: correct use of 'line of symmetry,' 'mirror line,' and 'congruent halves.' Misuse of 'equal' (meaning only equal length) instead of 'congruent' (meaning same shape AND size) is a diagnostic flag — address at the start of Lesson 3. Count the 'I still wonder' notes that anticipate Lesson 3 content (properties of reflection, distances from mirror line) — these show readiness to advance.
Model	Learners return to their exercise	Say: 'Every lesson we add to our	Cumulative Model Revision — the	Quickly scan 6–8 exercise books

Building Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Number Lines: Football Game (PLIX) Source: CK-12 🔍 Search ARES for similar readings	books and update their 'Unit Model' — a running annotated diagram that tracks the class's growing explanation of the matatu mirror phenomenon. In Lesson 1 they drew the matatu scenario with the lorry and its image and labelled: 'image same size as object,' 'image appears flipped,' 'image appears as far behind mirror as object is in front.' Today they ADD: (1) a dotted line on the mirror surface labelled 'line of symmetry / mirror line,' (2) an arrow on each side of the line labelled 'congruent half,' and (3) a small inset box listing the shapes explored today with their line-of-symmetry counts. Learners also sketch a Maasai beadwork collar and draw in at least two lines of symmetry, captioning: 'Symmetry in Kenyan culture — same mathematics as the matatu mirror.'	model of the matatu mirror. Today we know more than we did yesterday. Open your exercise books to your Unit Model diagram.' Project or draw on the board a sample update showing the dotted mirror line being added. Say: 'Add the dotted mirror line and label it. Then add your line-of-symmetry table as an inset box.' Give 4 minutes. [WAIT TIME: 4 minutes.] Circulate and check that: (a) the mirror line is drawn ON the mirror surface, not behind it; (b) both sides are labelled 'congruent half.' After 4 minutes, cold-call 2 learners to hold up their updated models for the class. [WAIT TIME: 10 seconds commentary per learner.] Close by saying: 'Your model today answers part of our driving question: the two halves match because the mirror line divides the figure into congruent mirror images. In our next lesson, we will investigate exactly HOW far each point of the lorry moves when it is reflected — and that will tell us even more about why the image is the same size.'	ongoing annotated diagram externalises the learner's mental model, making misconceptions visible and progress tangible. Requiring learners to add each lesson's new insight to the same diagram (rather than starting afresh) builds coherent schema for the full unit and mirrors the scientific practice of iterative model refinement.	to check model updates. Key indicators: mirror line drawn on the mirror surface (not in the air); both halves labelled 'congruent'; line-of-symmetry counts consistent with the class table from the Observe phase. Learners whose diagrams show the mirror line behind the mirror surface (a persistent spatial misconception) should be seated at the front during Lesson 3's paper-folding activity.

D. TEACHER REFLECTION

1. During the Observe phase, did all pairs successfully locate the correct number of lines of symmetry for the rhombus (2 lines along the diagonals)? If not, which alternative fold lines were learners attempting, and how can I address this misconception more explicitly in Lesson 3 using coordinate geometry?\n2. How confidently did learners articulate the connection between the line of symmetry and the matatu mirror line in their 'I now know' sticky notes? Were the connections specific (mentioning 'mirror line,' 'congruent halves') or vague — and what does this tell me about readiness for the formal definition of reflection in Lesson 3?\n3. Did the Maasai beadwork and kikoi context genuinely engage learners, or did it feel like a bolt-on addition? How might I integrate cultural symmetry patterns more deeply into the mathematical tasks themselves in future lessons?\n4. Were there learners who finished the paper-folding activity very quickly and appeared under-challenged? What extension task (e.g., 'Find a shape in this classroom that has NO line of symmetry and prove it') should I prepare for Lesson 3?\n5. How well did the 'I still wonder' notes anticipate the content of Lesson 3 (properties of reflection: equal distances from the mirror line, perpendicularity)? Will I need to seed those ideas myself in the opening of Lesson 3, or did learners generate them organically?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that placing a plane mirror along certain lines on a maize leaf, bottle cap, and paper shapes produces a reflection that completes the original figure perfectly — and that different shapes (equilateral triangle, rectangle, rhombus, regular hexagon) have different numbers of such lines. They also observed symmetry in Maasai beadwork and kikoi fabric designs.
What did I learn?	A line of symmetry divides a plane figure into two congruent halves, each of which is the exact mirror image of the other. An equilateral triangle has 3 lines, a rectangle has 2, a rhombus has 2 (along its diagonals), and a regular hexagon has 6. Some figures have one line, some have many, and some have none.
How does this explain the phenomenon?	The matatu mirror shows a perfect image of the lorry because the mirror surface acts as a line of symmetry — it divides the combined 'lorry + image' system into two congruent halves. This is WHY the image looks exactly like the lorry: the mirror line guarantees that one half (the image) is a perfect, congruent copy of the other half (the real lorry).

LESSON 3 (40 minutes): Reflection on the Cartesian Plane — Plotting Object and Image Coordinates

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners move from physical paper-folding to the Cartesian plane, plotting object and image coordinates under reflection across horizontal, vertical, and diagonal mirror lines — connecting measured distances to the matatu rear-view mirror phenomenon.
Knowledge	– State and apply the three key properties of reflection: image distance = object distance from the mirror line; the line joining object to image is perpendicular to the mirror line; the mirror line is the perpendicular bisector of the segment joining each object point to its image point. – Identify and write the equations of common mirror lines on the Cartesian plane (x-axis: $y = 0$; y-axis: $x = 0$; $y = x$; $y = -x$). – Determine the coordinates of the image of a point or polygon reflected across a given mirror line on the Cartesian plane.
Skills	– Plot object and image points accurately on a Cartesian plane using a ruler and geometrical set. – Apply the perpendicularity and equal-distance properties to locate image coordinates without folding.
Attitudes	– Appreciate how abstract coordinate rules emerge directly from physical observations made in earlier lessons. – Show responsibility and precision when constructing diagrams, recognising that inaccurate plots lead to incorrect conclusions about the mirror phenomenon.
Key Inquiry Question	How do the properties of reflection discovered physically in Lesson 2 translate into exact coordinate rules on the Cartesian plane — and how do those rules explain why the matatu mirror image is always the same size and shape as the lorry behind it?
Purpose in Storyline	In Lessons 1 and 2 learners established symmetry lines and discovered the physical properties of reflection through paper-folding and plane mirrors. Lesson 3 bridges that physical intuition to the formal Cartesian plane: learners encode the equal-distance and perpendicularity properties as coordinate rules, plot reflected polygons, and gather the numerical evidence needed to later prove that every reflected image is congruent to its object — directly answering why the lorry in the matatu mirror looks exactly the same size.
Safety Notes	Compasses and drawing pins have sharp points — remind learners to handle them carefully and place them on the desk when not in use. No other specific hazards.

B. LESSON OVERVIEW

This lesson is the third step in the evidence-gathering sequence for the matatu rear-view mirror phenomenon. In Lessons 1 and 2, learners identified lines of symmetry in plane figures and used physical mirrors and paper-folding to discover that a reflected image is as far behind the mirror line as the object is in front of it, and that the joining segment is perpendicular to the mirror line. Lesson 3 now asks: can we capture those physical properties as precise coordinate rules? Learners are given pre-drawn Cartesian grids and guided through reflecting a lorry-shaped polygon (a simple rectangle labelled to represent the Nairobi–Thika lorry) across four different mirror lines — the x-axis, the y-axis, $y = x$, and $y = -x$ — recording how each vertex coordinate transforms.

The core sensemaking activity is a structured investigation in pairs: learners plot an object point, count grid squares perpendicularly to the mirror line, count the same distance on the other side, and record the new coordinates. After four reflections they compare their coordinate tables and articulate a rule for each mirror line (e.g., reflection in the y-axis maps $(a, b) \rightarrow (-a, b)$). The lesson closes by returning to the phenomenon: because coordinates change in a predictable, rule-governed way, every length between any two vertices is preserved — exactly what makes the lorry image in the matatu mirror the same size as the real lorry.

By the end of the lesson learners will have generated a personal Reference Card of coordinate reflection rules, updated their Driving Question Board with new evidence, and revised their cumulative model to show mathematical notation alongside the physical arrows from Lesson 2. This sets up Lesson 4, where learners will determine the equation of an unknown mirror line given only the object and its image.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-page Cartesian grid (axes from −6 to 6) with a single point P(3, 2) marked. The teacher asks: 'Without drawing anything yet, write down in your notebook what you think the coordinates of P's mirror image will be if the mirror line is the y-axis.' Learners write a prediction individually (1 minute), then share with a shoulder partner. Two or three pairs share out to the class before any instruction.	Distribute grids as learners settle. Say: 'Look at the grid. You can see point P at (3, 2). We are going back to the matatu mirror — but now the mirror is a line on this grid. The y-axis is the mirror. Where will P's image appear? Write your prediction — coordinates only — in your notebook. You have 60 seconds. Go.' [WAIT — allow 60 seconds of silent writing.] After 60 seconds: 'Share your prediction with the person next to you. Do you agree or disagree? Why?' [WAIT 30 seconds for partner talk.] Cold-call 3 pairs: 'Pair at the front left — what did you predict and why?' Accept all answers without confirming or correcting. Write predicted coordinates on the board in a T-table headed 'Our Predictions.' Say: 'Keep these predictions visible — we will check them in the next phase.'	Prediction with Justification: asking learners to commit a coordinate pair to paper before instruction activates prior knowledge from Lesson 2 (equal distances from mirror line) and creates cognitive tension when predictions vary — motivating careful observation in the next phase.	Teacher scans notebooks during partner share: look for whether learners correctly negate the x-coordinate (−3, 2), reflect both coordinates (−3, −2), or make other errors. Tally the frequency of each prediction type on the board — this data guides how much time to spend on the y-axis rule in the Observe phase.
Observe Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry	Working in pairs, learners carry out a structured four-part grid investigation. They receive a Reflection Investigation Worksheet with four separate Cartesian grids, each showing a small lorry-shaped quadrilateral ABCD (vertices A(1,1), B(4,1), C(4,3), D(1,3)) and a different mirror line labelled on each grid: (i) y-axis (x = 0), (ii) x-axis (y = 0), (iii) y = x, (iv) y = −x. For each grid, learners: (a) draw the perpendicular from each vertex	Distribute the worksheet. Say: 'This quadrilateral represents the lorry that the matatu conductor is watching in the mirror. Your job is a mathematician's job: find the exact rule that the mirror follows. Work in pairs. Use your ruler and set-square every time — do not estimate.' [Circulate for 2 minutes, checking that learners are drawing perpendiculars and not just reflecting by eye.] After 2 minutes, pause the class: 'Quick check —	Structured Data Collection with Pattern Detection: the four-grid parallel structure allows learners to notice that each mirror line produces a different-but-systematic coordinate transformation. Recording raw coordinates before writing the rule mirrors the NGSS practice of Analysing and Interpreting Data — learners derive the abstraction from their own measurements rather than receiving it as a	Circulate and check: (1) Are perpendiculars genuinely perpendicular (set-square used)? (2) Are image vertices equidistant from the mirror line — teacher counts squares alongside learners when errors are spotted. (3) In the Coordinate Rule Table, watch for the common confusion on y = x: learners may write (a, b) → (−b, −a) instead of (b, a). Identify any pair making this error and address in the Explain phase.

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	to the mirror line using a ruler and set-square; (b) count grid squares from vertex to mirror line; (c) count the same number of squares on the other side; (d) plot and label the image vertex A′B′C′D′; (e) record the object and image coordinates in a table. After completing all four grids, learners complete a Coordinate Rule Table: they write the general rule (a, b) → (?, ?) for each mirror line.	hold up your set-square. Everyone should be using it to make a 90-degree angle with the mirror line. If you are not, adjust now.' [Resume circulation.] At 8 minutes, stop the class for a mid-activity share: 'Pairs who have finished grids (i) and (ii), tell me — what pattern do you see in the x-coordinates when you reflect across the y-axis?' Cold-call 2 pairs. [WAIT 10 seconds after each response before probing.] Say: 'Does the lorry on grid (i) look the same size as the original? Count the side lengths and tell me.' [WAIT 15 seconds.] Continue: 'Finish grids (iii) and (iv). You have 8 more minutes.' Circulate and prompt struggling learners: 'Draw your perpendicular first. How far is vertex A from the line y = x? Count diagonally along the grid squares.' At 16 minutes, bring class back together. Cold-call 3 different pairs to read their coordinate rule for each of the four mirror lines. Record the rules on the board in a class Reference Table.	definition.	
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually complete a Reflection Rules Reference Card — a half-page template with four rows (one per mirror line) where they write the mirror line equation, the coordinate rule, and a worked example using one vertex of their lorry quadrilateral. They then answer two consolidation questions: (Q1) A point K(−2, 5) is reflected across the line y = x. Write the coordinates of K′. (Q2) The image of point M after reflection is M′(4, −3). If the mirror line is the x-axis, write the coordinates of M. After individual work (5 minutes), pairs compare answers and resolve	Say: 'You have discovered four coordinate rules — now prove you own them. Complete your Reference Card individually. No partner help for the first 4 minutes.' [Set visible timer. Circulate silently for 4 minutes, noting errors on y = x and y = −x.] After 4 minutes: 'Compare your Reference Card with your partner. If you disagree, use the grid to check — go back to counting squares.' [WAIT 2 minutes for partner reconciliation.] Facilitate whole-class discussion: 'Let's go back to the matatu. The lorry is 3 metres wide. Its left headlight is at position L and right headlight at R. After reflection in	Worked Example with Backward Question: Q2 reverses the usual direction (giving the image, asking for the object), deepening understanding that reflection is its own inverse — a key property that learners will need when determining mirror-line equations in Lesson 4. Connecting the distance-preservation result explicitly to the phenomenon completes the evidence loop begun in Lesson 1.	Collect Reference Cards at the end of the lesson and scan for: (1) Correct rule for all four mirror lines — especially y = x: (a, b) → (b, a). (2) Correct answer to Q1: K′(5, −2). (3) Correct answer to Q2: M(−4, −3) — wait, M′(4,−3) reflected in x-axis means M(4, 3). Check that learners correctly identify that reflecting the image back across the same mirror line returns the original point. Use errors to plan the opening review of Lesson 4.

	disagreements. The teacher facilitates a whole-class discussion connecting coordinate rules back to the phenomenon: 'Now that we have these rules, can we explain mathematically why the lorry in the matatu mirror is the same size as the real lorry?'	the mirror line, where are L′ and R′?' Cold-call 2 learners. [WAIT 12 seconds after each.] Then ask: 'The distance from L to R — is it the same as from L′ to R′? How do your coordinate rules tell you that?' [WAIT 15 seconds.] Guide learners to articulate: 'Because the rule maps (a, b) → (−a, b), the horizontal distance between any two points is preserved — just flipped in sign. That is why the lorry image is the same width as the lorry.' Write on the board: 'Reflection preserves distance → image and object are the same size and shape.' Say: 'Write this sentence in a box on your Reference Card.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each pair writes two sticky notes: one GREEN note headed 'What we now know' and one ORANGE note headed 'New question.' The green note must contain a coordinate rule discovered today plus a sentence connecting it to the matatu mirror phenomenon. The orange note must ask a question that today's lesson raised but did not answer — e.g., 'What if we only have the object and image — how do we find the mirror line equation?' Pairs post their notes on the class DQB under the columns 'Evidence Gathered' and 'Still Wondering.' The teacher reads three orange notes aloud and frames them as the target for Lesson 4.	Distribute sticky notes (or learners write on paper squares). Say: 'Two minutes — write your green note first. It must include a coordinate rule AND connect it to the lorry in the matatu mirror. Then write your orange note — what do you still want to know?' [WAIT 2 minutes.] 'Post your notes on the board.' [Allow 1 minute for posting.] Read three orange notes aloud: 'I am hearing questions like: How do we find the mirror line if we only see the object and image? Can the mirror line be a slanted line that is not y = x? Does reflection always make the image congruent? These are exactly the questions real mathematicians ask.' Circle the orange notes that relate to finding the mirror-line equation and say: 'We will answer these in our next lesson.'	Two-Column DQB Update (Evidence / Still Wondering): separating confirmed knowledge from new questions makes the cumulative nature of the inquiry visible. The physical act of posting notes gives every learner a public stake in the investigation and models scientific practice of tracking what is known versus what remains to be explained.	Read all posted green notes quickly: check that learners are connecting coordinate rules (not just listing them) to the phenomenon. Flag any green notes that state only 'reflection maps (a,b) to (−a,b)' without a phenomenon link — prompt those pairs verbally: 'How does that rule explain what the matatu driver sees?'
Model Building Phase 📹 **VIDEO:**	Learners retrieve their cumulative model from Lesson 2 (a hand-drawn diagram showing the matatu mirror setup with arrows	Say: 'Take out your model from last lesson. You are going to upgrade it with today's mathematics.' [Give 4 minutes for	Cumulative Model Revision (Progressive Schematisation): adding mathematical notation (coordinate rules, mirror-line	Examine 4–5 models during circulation: look for (1) correct placement of image vertices using today's coordinate rules; (2) the

Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	indicating equal distances and perpendicularity). They now add a Cartesian overlay: they sketch coordinate axes onto their diagram, place the lorry quadrilateral at approximate grid coordinates, label the mirror line with its equation, and annotate the image vertices with their transformed coordinates. They also add a 'Distance Preservation Statement' arrow pointing from the object to the image with the sentence: 'All side lengths are preserved because reflection maps distances symmetrically across the mirror line.' Learners write one sentence below their model: 'This tells us that the image is always [same size / different size] and [same shape / different shape] as the object, which means they are ___________.' (The blank will be filled with 'congruent' once the formal congruence tests are introduced in Lessons 5–7.)	model revision. Circulate and prompt: 'Can you add the equation of the mirror line? Where are the image coordinates? Can you show why the side lengths are the same?'] After 4 minutes: 'Look at the sentence at the bottom of your model — the one with the blank. Do not fill in the blank yet. We are gathering evidence to earn that word. Keep your model safe — you will add to it in every lesson.' Cold-call 2 learners to share their upgraded model with the class. Ask the class: 'Does the matatu driver's mirror show an image that follows these coordinate rules? Which mirror line best represents a flat rear-view mirror?' [WAIT 15 seconds.] Accept responses — guide learners to see that a flat rear-view mirror is best modelled as a vertical line (x = k for some constant k), which maps (a, b) → (2k − a, b), preserving the y-coordinate (height) and reflecting the x-coordinate — explaining why the lorry appears at the same height but laterally flipped.	equations) to a diagram that began as a physical sketch makes the transition from concrete to abstract visible and trackable. The deliberate blank ('congruent') creates productive anticipation — learners know the model is incomplete, motivating continued engagement across the unit.	mirror-line equation written on the diagram; (3) the distance-preservation annotation. Use models that correctly show all three as exemplars to share at the start of Lesson 4. Note models that still show reflected images on the wrong side of the mirror line — these learners need a brief one-to-one check at the start of Lesson 4.

D. TEACHER REFLECTION

1. During the Observe Phase, did all pairs use a set-square to draw perpendiculars to the mirror line, or were some learners reflecting by visual estimation? What adjustment would improve precision next time — for example, pre-printed perpendicular guide dots on the worksheet?\n2. The rule for reflection across y = x (swapping coordinates) is consistently the hardest for learners. Which instructional move — counting diagonal grid squares, folding the paper, or a colour-coded table — helped the most today, and how will I reinforce it at the start of Lesson 4?\n3. Did the matatu-mirror context remain alive in learners' minds throughout the Explain Phase, or did it fade once the coordinate algebra began? What specific linking question could I insert to re-anchor the mathematics to the phenomenon?\n4. Looking at the DQB orange notes: did learners' questions genuinely point toward finding the mirror-line equation (the content of Lesson 4), or did they reveal unexpected gaps (e.g., confusion about negative coordinates) that I need to address before moving on?\n5. How many learners correctly completed the 'blank' sentence at the bottom of the model with reasoning tied to distance preservation, even without yet knowing the word 'congruent'? This gives me a baseline for how ready the class is for the formal congruence tests in Lessons 5–7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners plotted the vertices of a lorry-shaped quadrilateral and its reflected images across four mirror lines (x-axis, y-axis, $y = x$, $y = -x$) on Cartesian grids, observing that each image vertex is exactly as far from the mirror line as the original vertex, that the joining line is perpendicular to the mirror line, and that all side lengths of the image equal the corresponding side lengths of the object.
What did I learn?	Each mirror line produces a specific, predictable coordinate rule: reflection in the y-axis maps $(a, b) \rightarrow (-a, b)$; in the x-axis maps $(a, b) \rightarrow (a, -b)$; in $y = x$ maps $(a, b) \rightarrow (b, a)$; in $y = -x$ maps $(a, b) \rightarrow (-b, -a)$. These rules preserve all distances between points, meaning the image is always the same size and shape as the object.
How does this explain the phenomenon?	Because coordinate reflection rules preserve every distance between vertices, the lorry image in the matatu rear-view mirror must be exactly the same size and shape as the real lorry behind the vehicle — the coordinates change in a systematic (flipped) way, but no length is stretched or shrunk. This is the mathematical reason the conductor can use the mirror to judge the lorry's true size and distance, and it is the first precise evidence that the object and its mirror image may be congruent.

LESSON 4 (40 minutes): Properties of Reflection — Discovering the Four Rules of the Mirror Line	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use a plane mirror, tracing paper, and drawn arrow-shaped objects to discover and verify the four fundamental properties of reflection, directly explaining why the matatu rear-view mirror shows an image that is the same size as the lorry yet appears equidistant behind the glass.
Knowledge	– State the four properties of reflection: (i) the image is the same size as the object; (ii) the mirror line is the perpendicular bisector of the segment joining each object point to its corresponding image point; (iii) the image is laterally inverted; (iv) object and image are equidistant from the mirror line. – Explain why the image in a plane mirror appears as far behind the mirror as the object is in front of it. – Distinguish between the object distance and image distance measured perpendicularly to the mirror line.
Skills	– Measure and verify that corresponding object and image points are equidistant from the mirror line using a ruler and set square. – Use tracing paper folding along the mirror line to confirm that the object and its reflected image coincide exactly.
Attitudes	– Appreciate that mathematical properties of reflection underpin everyday road-safety tools such as the matatu rear-view mirror. – Show responsibility by handling the plane mirror carefully and recording measurements with integrity.
Key Inquiry Question	How do we use reflection in day-to-day life?
Purpose in Storyline	Having plotted reflection images on the Cartesian plane in Lesson 3, learners now move from coordinate rules to physical discovery: they use a real plane mirror and tracing paper to derive — rather than be told — the four properties of reflection, giving them the geometric reasoning needed to explain why the lorry in the matatu mirror appears exactly as far behind the glass as it truly is in front.
Safety Notes	Handle the small plane mirror with care — edges may be sharp. Do not shine the mirror into another learner's eyes. If a mirror breaks, do not touch the fragments; report to the teacher immediately.

B. LESSON OVERVIEW

This lesson is the pivotal evidence-gathering session in the unit. In Lessons 1–3 learners established lines of symmetry, explored reflection intuitively, and practised coordinate rules on the Cartesian plane. Now, in Lesson 4, they shift from rule-following to rule-discovery. Working in groups of four, each group receives a small plane mirror, a sheet of plain paper with a drawn arrow (representing the matatu from the anchoring phenomenon), a ruler, a set square, and tracing paper. Through a structured investigation they will independently discover — and then verify — each of the four properties of reflection. The lesson is grounded throughout in the matatu mirror phenomenon: the arrow on paper is explicitly framed as the lorry on the Nairobi–Thika highway, and the mirror line is the rear-view mirror glass.

The lesson follows the five-phase ARES framework. In the Predict phase, learners commit to written predictions about what the mirror image of the arrow will look like relative to the mirror line — size, distance, and orientation. In the Observe phase, groups place the mirror on a drawn line, observe the image, mark image points, remove the mirror, and measure. In the Explain phase, groups use tracing paper folding to verify coincidence and articulate each property in their own words before a class consolidation. The Driving Question Board is updated with new evidence cards, and in the Model Building phase learners annotate their cumulative class diagram with perpendicular bisector marks and equal-distance arrows.

By the end of the lesson every learner should be able to state the four properties of reflection with confidence, connect Property IV (equidistance) directly to the matatu phenomenon, and recognise that the mirror line is always the perpendicular bisector of any object-image connector segment — a geometric fact that will anchor their work on finding mirror-line equations in Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Identify and Apply Number Properties in Integer Operations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-sheet with a sketch of the anchoring phenomenon — a lorry approaching on the Nairobi–Thika highway and a dashed line labelled 'mirror surface.' Beneath the sketch are four prediction prompts: (1) Will the image of the lorry be larger, smaller, or the same size as the real lorry? (2) Will the image appear closer to, farther from, or the same distance from the mirror as the real lorry? (3) Will the image face the same direction as the lorry or be flipped? (4) If you drew a line connecting the lorry's left headlight to its image's left headlight, what angle would that line make with the mirror surface? Learners write individual responses in pencil (2 minutes), then share with a shoulder partner (1 minute). Groups record a consensus prediction on a sticky note to be displayed.	Display the anchoring phenomenon image on the board. Say: 'Last lesson we used coordinate rules to predict where an image lands on the Cartesian plane. Today we go back to the matatu on the Nairobi–Thika highway. Before we pick up any mirror or ruler, I want you to commit to a prediction — in writing, in pencil — because a good scientist writes down what they think before they look.' Distribute the prediction sheets. After 2 minutes, say: 'Turn to your shoulder partner — the person sitting next to you — and compare your answers. Do you agree or disagree? Be ready to defend your prediction.' Allow 1 minute. Then cold-call exactly 3 groups: 'Group 2 — what did you predict for the distance question?' [WAIT 12 seconds.] 'Group 5 — same question, do you agree or disagree with Group 2, and why?' [WAIT 12 seconds.] 'Group 7 — what about the angle of the connecting line?' [WAIT 15 seconds.] Do NOT confirm or correct predictions yet. Post all sticky notes on the board under the heading 'Our Predictions.'	Prediction Commitment (Elicit–Commit–Reveal): By writing predictions before observing, learners activate prior knowledge and create cognitive dissonance that sharpens attention during the observation phase. Anchoring every prompt to the lorry/mirror phenomenon ensures predictions are not abstract — they carry the emotional hook of a real road context learners recognise from matatu journeys.	Teacher scans written predictions as they circulate. Look for: (a) whether learners predict 'same size' or erroneously predict size change — a residual misconception from Lesson 1 enlargement work; (b) whether any learner already states 'equal distance' or 'perpendicular' — flag these learners to share during Explain phase. Record which groups predict 90° for the angle; this reveals readiness for the perpendicular bisector property.
Observe Phase 📹 **VIDEO:** [Proofs with	Each group of four receives a materials tray: one small plane mirror (held vertically using a lump of Blu-Tack), one A4 sheet with a	Before groups begin, teacher models Step 2 slowly at the board: 'This is the trickiest part — marking the image point through the mirror.	Data-Table Evidence Collection with Physical Verification: Learners generate their own quantitative evidence (six distance	Teacher checks results tables at each group for two indicators: (i) Do corresponding distances match (e.g., distance from A to mirror line

transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identify and Apply Number Properties in Integer Operations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	bold arrow drawn on it (the 'matatu arrow' — tip labelled A, tail labelled B, a midpoint labelled M), a ruler, a set square, a sharp pencil, and a sheet of tracing paper. Step 1: Groups draw a straight horizontal mirror line across the middle of the paper, stand the mirror on it, and observe the image of the arrow in the mirror. Each learner sketches what they see and labels the image tip A', tail B', midpoint M'. Step 2: Groups mark each image point by placing a pencil dot through the mirror at A', B', and M' (the teacher demonstrates this technique: hold the mirror firmly, look at the image, and touch the mirror surface directly behind the image point with the pencil tip). Step 3: Groups remove the mirror, join A to A', B to B', and M to M' with straight lines. Step 4: Using a ruler, groups measure (i) the distance from A to the mirror line, and the distance from A' to the mirror line; (ii) same for B and B'; (iii) same for M and M'. Groups record all six measurements in a results table. Step 5: Using a set square, groups check whether each connecting segment (AA', BB', MM') is perpendicular to the mirror line and record 'Yes' or 'No.'	Watch: I place my pencil tip right behind where I see the image of point A in the mirror. I am not guessing — I am using the image I can see to guide my pencil. Everyone watch before you try.' Circulate continuously during the activity. At the 5-minute mark, prompt groups that are stuck on Step 2: 'Close one eye and align your pencil directly with the image you see.' At the 8-minute mark, prompt measurement recording: 'Have you filled in all six distances in your table? Use your ruler — measure in millimetres for precision.' Ask probing questions at each table (not whole-class): 'What do you notice about the two distances on either side of the mirror line for point A?' [WAIT 10 seconds for group discussion before moving on.] 'Is your AA′ line crossing the mirror line at a right angle? Use your set square to check — don't guess.' If a group finishes early: 'Now try with a diagonal mirror line — draw it at 45°. Do your results still hold?' Alert all groups at 12 minutes: 'You should be at Step 5 now — recording your perpendicularity check.'	measurements) rather than reading values from a textbook. The dual verification — ruler measurement AND set-square perpendicularity check — mirrors scientific practice (NGSS Science and Engineering Practice 3: Planning and Carrying Out Investigations) and makes each property a discovered fact rather than a received rule. Connecting the arrow to the 'matatu' explicitly keeps the phenomenon alive during data collection.	= distance from A' to mirror line, within ±2 mm measurement error)? Groups with large discrepancies have likely marked image points inaccurately — teacher assists with the mirror-marking technique. (ii) Do all groups record 'Yes' for perpendicularity? Groups recording 'No' have likely drawn connecting lines incorrectly — teacher checks their diagram. Flag any group whose table shows consistent equal distances: 'Hold that result — you will share it with the class in Explain phase.'
Explain Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identify and Apply Number Properties in	Groups now fold their tracing paper exactly along the mirror line and observe whether the object arrow lands precisely on the image arrow — verifying that the image is the same size and shape (congruent) as the object. Each group then completes a 'Properties Card' with four sentence starters: (i) 'The size of the image is _____ because _____.'; (ii) 'The distance from the object to the mirror line is _____ the distance from the image	Initiate the tracing-paper fold: 'Fold your tracing paper exactly along the mirror line — crease it firmly. Now hold it up to the light. Does the arrow on one half land exactly on top of the arrow on the other half?' [WAIT 10 seconds for groups to observe.] 'What does that tell you?' [WAIT 12 seconds.] Cold-call 2 groups: 'Group 3, what did you see when you folded?' [WAIT 10 seconds.] 'Group 6, do you agree? What word in	Properties Card + Gallery Walk (Claim–Evidence–Reasoning): The sentence-starter format scaffolds scientific argumentation by requiring learners to state a claim, cite their measured evidence, and provide a reason — matching NGSS Practice 6 (Constructing Explanations). The gallery walk distributes the sensemaking across all groups rather than centering it in the teacher, building collaborative mathematical	Teacher reads completed Properties Cards during the gallery walk, checking for: (i) Correct use of 'equal' or 'equidistant' for Property IV; (ii) Use of '90 degrees' or 'perpendicular' for Property III; (iii) Use of 'congruent' or 'same size' for Property I; (iv) Use of 'laterally inverted,' 'flipped,' or 'reversed' for Property III. Any group that writes 'the image is smaller' or 'closer to the mirror' holds a persistent misconception

Integer Operations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	to the mirror line because _____.'; (iii) 'The line joining an object point to its image point meets the mirror line at _____ degrees because _____.'; (iv) 'When I look at the image in the mirror, the arrow points _____ compared to the object because _____.'. Groups share their completed cards in a class gallery walk (each card posted on the wall), then the teacher leads a 5-minute whole-class consolidation to formalise all four properties, connecting each explicitly to the matatu lorry.	mathematics describes two shapes that are exactly the same size and shape?' [WAIT 12 seconds — accept 'congruent' and write it on the board.] During gallery walk, circulate and prompt: 'Read Group 4's card — do you agree with how they described Property III? Why or why not?' For the consolidation, write the four properties on the board ONE AT A TIME, each time asking: 'Which part of our lorry-mirror story does this explain?' For Property IV (equidistance), say: 'This is the one the matatu conductor relied on — if the lorry is 20 metres in front of the mirror glass, where does its image appear to be?' [WAIT 15 seconds.] Cold-call 1 learner: 'Akinyi, tell us.' Confirm: '20 metres behind the mirror glass. That is Property IV — equidistance. That is why the rear-view mirror is useful: the image tells the driver exactly how far back the lorry truly is.'	discourse. Tying each property back to the lorry phenomenon ensures transfer from abstract to applied understanding.	— teacher addresses individually during consolidation. Success criterion: at least 80% of groups correctly complete all four sentence starters before the consolidation.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identify and Apply Number Properties in Integer Operations (Flexbook) Source: CK-12	Each group receives two colour-coded cards: a GREEN 'We Now Know' card and an ORANGE 'We Still Wonder' card. On the green card, groups write one statement that their evidence from today's investigation has confirmed about the matatu mirror — connecting directly to the driving question ('Why does the rear-view mirror show an image that is the same size yet appears flipped?'). On the orange card, groups write one new question raised by today's work (e.g., 'If the mirror line is diagonal, do we use the same rule?' or 'Can we find the equation of any mirror line from just the object and image points?'). Cards are posted on the class Driving Question Board	Direct learners to the DQB: 'Before the next lesson, we update our board. Remember our driving question — it is written at the top. Today's investigation gave us real evidence. On your green card, write ONE thing we now know for certain, using today's measurements as your evidence. On your orange card, write ONE new question that today's work made you wonder about.' Allow 3 minutes for writing. Collect and post cards. Read 2 green cards aloud: 'Listen to what Group 1 wrote: 'The mirror line is exactly in the middle — at equal distances from the lorry and its image.' Does everyone agree that our measurements prove that?' [WAIT	Driving Question Board Update (Claim Mapping): Colour-coded cards make the distinction between settled knowledge and open questions visible and permanent. Posting the cards publicly creates accountability — learners know their green-card claims must be evidence-backed. Drawing explicit arrows from cards to the driving question models the scientific practice of connecting evidence to explanatory frameworks (NGSS Practice 8: Obtaining, Evaluating, and Communicating Information). The orange cards seed Lesson 5's inquiry into mirror-line equations.	Teacher reads all green cards before the next lesson. Success indicator: green cards should reference at least one of the four properties and connect it to the lorry/mirror phenomenon. Red flag: if green cards say only 'reflection flips the image' without mentioning equidistance or perpendicularity, the Explain phase consolidation was insufficient and the teacher should open Lesson 5 with a brief recap of Properties II and IV.

🔍 Search ARES for similar readings	under the columns 'What we now know' and 'New questions.' The teacher reads selected cards aloud and draws explicit arrows on the DQB connecting today's green cards to the driving question.	10 seconds.] Then say: 'And Group 4 is still wondering: 'Can we write an equation for the mirror line if we only know the object and image coordinates?' — that is exactly what Lesson 5 is about. Keep that question on the board; we will answer it next time.' Draw a dotted arrow from Group 4's orange card to the driving question. Summarise: 'Our DQB is growing. We now know four properties. Our driving question asks about size and flipping — we have answered both parts with evidence today.'		
Model Building Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Identify and Apply Number Properties in Integer Operations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative class diagram that has been building since Lesson 1 — a large A3 poster at the front of the room showing the matatu on the Nairobi–Thika highway with the mirror, the lorry as object, and space for the image. Today, each learner adds to their own notebook version of this diagram: they draw the lorry arrow as the object, draw the mirror line, draw the image arrow on the other side, and then annotate the diagram with four labelled features: (i) a double-headed arrow showing equal distances from the lorry tip to the mirror line and from the image tip to the mirror line, labelled 'd = d''; (ii) a small square symbol at the point where the connector AA′ meets the mirror line, showing perpendicularity; (iii) a curved arrow showing lateral inversion; (iv) tick marks on the object and image arrows showing equal length. Learners write a one-sentence caption beneath their diagram: 'The mirror line is the perpendicular bisector of every segment joining an object point to	Distribute learners' cumulative notebooks or direct them to the relevant page. Say: 'Our model of the matatu mirror has been growing each lesson. Lesson 1 — we added lines of symmetry. Lesson 2 — we added the concept of reflection. Lesson 3 — we added coordinate rules. Today we add the four properties — the evidence we discovered ourselves.' Display a completed annotated diagram on the board as a reference (not a copy-sheet — learners should draw their own). Circulate and check that every learner includes all four annotations. Prompt individuals: 'Show me where you have marked equal distances. Is your square symbol at the correct point — where the connector meets the mirror line, not somewhere along it?' At the 3-minute mark, ask the class to hold up their notebooks: 'Look at your neighbour's diagram. Do they have the perpendicularity symbol? Do they have the equal-distance arrows? Give them a thumbs-up if their diagram is complete.' Close the phase by	Cumulative Model Annotation (Progressive Model Revision): The cumulative diagram functions as a living scientific model that learners revise with each new lesson's evidence — mirroring how scientists update models as new data emerges (NGSS Practice 2: Developing and Using Models). Requiring four distinct annotations rather than a single drawing forces learners to represent each property separately, deepening encoding. The one-sentence caption bridges visual and verbal representations of the perpendicular bisector property, which is the conceptual lynchpin for Lesson 5.	Teacher checks notebooks during circulation for all four annotations present and correctly placed. Specific indicators: (i) equal-distance arrows must span from object point to mirror line AND from mirror line to image point — not from object to image; (ii) perpendicularity square must sit at the intersection point on the mirror line; (iii) lateral inversion arrow must show direction reversal, not size change. Any missing annotation is corrected before the learner closes their notebook. Exit criterion: every learner's diagram has all four features labelled and the one-sentence caption written.

	its image point.'	pointing to the class poster: 'I am going to add today's evidence to our class model right now — watch.' Add the four annotations to the class poster in front of learners. Say: 'Next lesson we will use these four properties to find the equation of any mirror line — we will finally be able to answer Group 4's orange card question.'		

D. TEACHER REFLECTION

1. During the Observe phase, were all groups able to accurately mark image points through the mirror? If two or more groups struggled with the mirror-marking technique, consider beginning Lesson 5 with a brief guided repeat of this step before moving to mirror-line equations.
2. In the Explain phase, did learners spontaneously use the word 'congruent' or did it require teacher prompting? If it required prompting, plan an explicit vocabulary warm-up in Lesson 5 connecting 'congruent' to the four properties discovered today.
3. Did the distance measurements in learners' results tables show consistent equality (within ±2 mm), or were there systematic errors suggesting measurement technique problems? If errors were systematic, consider whether the mirror-marking method needs a classroom demonstration video or a physical jig.
4. Were the orange DQB cards focused productively on mirror-line equations (anticipating Lesson 5), or did they reveal deeper misconceptions about what reflection means? Read all orange cards before planning Lesson 5 and adjust the opening accordingly.
5. Did the tracing-paper folding activity convincingly persuade all learners that the object and image are congruent, or did some groups express doubt? If doubt persists, plan a paper-cutout congruence activity as a supplementary task at the start of Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using a plane mirror placed on a drawn mirror line, learners observed that the image of the matatu arrow appeared the same size as the object, appeared to be the same distance behind the mirror line as the object was in front of it, and appeared laterally flipped. Measuring the distances from object points and image points to the mirror line confirmed equal distances, and a set-square check confirmed that the connecting segments met the mirror line at 90°. Folding tracing paper along the mirror line showed that the object and image coincide exactly.
What did I learn?	Reflection has four properties: (i) the image is the same size as the object; (ii) object and image are equidistant from the mirror line; (iii) the segment joining any object point to its corresponding image point is perpendicular to the mirror line — making the mirror line the perpendicular bisector of that segment; (iv) the image is laterally inverted relative to the object.
How does this explain the phenomenon?	These four properties explain why the lorry in the matatu rear-view mirror appears as far behind the mirror glass as the actual lorry is in front of it (Property II — equidistance), why it appears the same size as the real lorry (Property I — equal size, confirmed by the tracing-paper fold showing congruence), and why the lorry's right headlight appears on the left in the mirror (Property IV — lateral inversion). The mirror line — the glass surface — is always the perpendicular bisector of every object-image connector, making the image's position completely predictable.

LESSON 5 (40 minutes): Finding the Mirror Line — Determining the Equation of the Mirror Line Given an Object and Its Image	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners work backwards from a reflected image to determine the equation of the mirror line, deepening their understanding of the four properties of reflection established in Lesson 4.
Knowledge	– The mirror line is the perpendicular bisector of the segment joining any object point to its image point. – The midpoint of any object–image segment lies exactly on the mirror line. – The gradient of the mirror line is the negative reciprocal of the gradient of the object–image segment. – The equation of the mirror line can be derived using the midpoint and gradient of the perpendicular bisector.
Skills	– Calculate the midpoint of a segment joining an object point to its image point on the Cartesian plane. – Determine the gradient of the perpendicular bisector and write the equation of the mirror line in the form $y = mx + c$ or $x = k$ or $y = k$.
Attitudes	– Appreciate that reflection is a precise, rule-governed transformation and that hidden information (the mirror line) can always be recovered from observable evidence. – Show persistence and accuracy when working through multi-step coordinate geometry problems.
Key Inquiry Question	How do we determine the equation of the mirror line given an object and its image?
Purpose in Storyline	In previous lessons learners drew images given a mirror line; now they reverse the process — just as a traffic engineer might ask 'where exactly is the matatu's mirror positioned?' given only the object and its image — building the algebraic toolkit needed to prove congruence in Lesson 6.
Safety Notes	No specific hazards. If small hand mirrors are used for demonstration, remind learners to handle glass edges carefully and to avoid directing reflected sunlight into anyone's eyes.

B. LESSON OVERVIEW

This lesson marks a pivotal turn in the evidence-gathering sequence: instead of drawing the image given the mirror line, learners now reconstruct the mirror line given the object and its image. The lesson is anchored in the matatu phenomenon — if a road-safety inspector photographs both a lorry and its reflection in the rear-view mirror and needs to determine the exact position (equation) of the mirror surface, how would they do it mathematically? This motivating question drives the entire 40-minute period.

Building directly on Lesson 4's four properties of reflection — equal distances from mirror line, perpendicular connector, same size/shape, lateral flip — learners discover that the mirror line is always the perpendicular bisector of the segment joining each object point to its corresponding image point. They use coordinate geometry skills (midpoint formula, gradient, negative reciprocal gradient) already familiar from earlier strands to write the equation of the mirror line. Worked examples progress from simple horizontal/vertical mirror lines to oblique lines, including the special cases $y = x$ and $y = -x$ encountered in Lesson 3.

The lesson closes with learners updating the class Driving Question Board and revising their cumulative models. By the end of the lesson learners can state that: (i) the midpoint of any object–image pair lies on the mirror line; (ii) the mirror line is perpendicular to the object–image connector; and (iii) these two facts together uniquely determine the equation of the mirror line — a result that will directly support the congruence proofs in Lessons 6 and 7.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Use the Percent Equation to Find Part a (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a large Cartesian grid (drawn on the chalkboard or projected) with a single point P(2, 1) and its image P′(6, 5) marked, but with NO mirror line drawn. Working individually in their exercise books for 2 minutes, each learner sketches where they think the mirror line must be and writes one sentence justifying their prediction using the properties of reflection from Lesson 4. They then share predictions in a think-pair-share with their neighbour before a whole-class cold-call.	Display the grid and say: 'A matatu mirror has reflected point P(2, 1) to P′(6, 5). The mirror itself has been removed. In your book, draw where you think the mirror line was and explain why using what we discovered last lesson.' Allow 2 minutes of silent individual work — do NOT prompt further. After 2 minutes say: 'Turn to your neighbour. Compare your mirror lines. Do they agree? Why or why not?' Allow 1 minute of paired talk. Then cold-call THREE pairs: 'Akinyi and Otieno — show us your line'; 'Wanjiku and Kamau — do you agree?'; 'Fatuma and Hassan — what property from Lesson 4 guided you?' After each response allow 10 seconds of WAIT TIME before affirming or probing. Record 2–3 different predicted mirror lines on the board without evaluating them yet. Say: 'Let us investigate which prediction is correct — and more importantly, how we can be certain.'	Activation of Prior Knowledge via Predict-Before-Instruction: Learners retrieve the four properties of reflection from Lesson 4 and apply them predictively. This surfaces alternative conceptions (e.g., some learners may draw the mirror line through P′ rather than between P and P′) and creates cognitive need for the algebraic method.	Teacher scans books during the 2-minute silent work period, looking for: (1) whether learners position the mirror line between P and P′ (not beyond either point); (2) whether any learner correctly identifies the midpoint (4, 3) as lying on the mirror line; (3) quality of written justifications referencing 'equidistant' or 'perpendicular'. Learners who draw the line through P or P′ are flagged for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Use the Percent Equation to Find Part a (Flexbook)	Learners receive a structured worksheet (or copy a guided activity from the board) with three tasks. Task 1: Plot P(2,1) and P′(6,5) on a Cartesian plane, draw the segment PP′, find its midpoint M by folding/measuring, and mark M on the grid. Task 2: Using a set-square or the corner of their exercise book, draw a line through M that is perpendicular to PP′ — this is the proposed mirror line. Task 3: Verify the line is correct by checking that M is equidistant from	Distribute the worksheet or write the three tasks clearly on the board. Say: 'You are now a road-safety engineer. You have the lorry's position P and its mirror image P′. Your job is to find exactly where the mirror surface is.' Circulate during Task 1; after 3 minutes prompt: 'Has everyone found the midpoint? Remember the midpoint formula from your earlier work: $M = ((x_1+x_2)/2\ ,\ (y_1+y_2)/2)$.' Write the formula on the board. Circulate during Task 2; say	Guided Inquiry with Structured Recording: The tabular data-collection format helps learners see the pattern across multiple point pairs (extended in the Explain phase) rather than treating each calculation as isolated. The physical act of drawing PP′, locating M, and constructing the perpendicular at M connects the abstract formula to the geometric reality of the mirror, directly linking back to the phenomenon.	Teacher checks that: (1) all learners correctly compute M = (4, 3); (2) learners calculate gradient of PP′ = 1 and gradient of mirror line = −1; (3) learners can articulate the perpendicularity relationship using '$m_1 \times m_2 = -1$'. A common error to watch for: learners reversing the negative reciprocal (writing gradient = 1 instead of −1). If more than 4 learners make this error, pause the class for a 60-second whole-class clarification before moving on.

Source: CK-12 🔍 Search ARES for similar readings	P and P′ and that the angle at M is 90°. Learners record all measurements and observations in a table: Object Point | Image Point | Midpoint | Gradient of PP′ | Gradient of Mirror Line.	to the class: 'Before you draw your perpendicular, work out the gradient of PP′ first. What is rise over run for P to P′?' Cold-call TWO learners for the gradient calculation: 'Chebet — what gradient did you get for PP′? Show the class.' [Expected: (5−1)/(6−2) = 4/4 = 1.] Allow 10 seconds WAIT TIME. Then: 'Mwangi — if the gradient of PP′ is 1, what must the gradient of the perpendicular mirror line be?' [Expected: −1.] Allow 10 seconds WAIT TIME. During Task 3, ask: 'How does your verification connect to the matatu mirror? If the mirror line is wrong, what would happen to the driver's view of the road behind?'		
Explain Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Use the Percent Equation to Find Part a (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Using the midpoint M(4, 3) and gradient −1 found in the Observe phase, learners derive the equation of the mirror line step-by-step using $y - y_1 = m(x - x_1)$: $y - 3 = -1(x - 4) \rightarrow y = -x + 7$. They then verify: substitute P(2,1) — distance from line — and P′(6,5) — distance from line — to confirm equal distances. A second, more challenging example is then worked: A(−1, 4) reflected to A′(3, 0). Learners independently find the mirror line equation. The class closes this phase by generalising the three-step algorithm: (1) Find midpoint M. (2) Find gradient of AA′, then take negative reciprocal for mirror line gradient. (3) Use point-slope form to write the equation.	Bring the class back together at the board. Say: 'We have M(4,3) and gradient −1. How do we write the equation of this line? Think back to your coordinate geometry work.' Cold-call ONE learner to come to the board and write the derivation: 'Njeri — come and show us $y - y_1 = m(x - x_1)$ with our values.' Allow 15 seconds WAIT TIME while Njeri writes. If she stalls, prompt: 'Start by substituting m = −1 and the point (4, 3).' After Njeri finishes, ask the class: 'Thumbs up if you agree with y = −x + 7. Thumbs sideways if unsure.' Address any thumbs-sideways before moving on. Then say: 'Now let us verify this is truly the mirror line. Take point P(2,1). Can you show that P is the same distance from the line y = −x + 7 as P′(6,5) is?' Guide learners through the perpendicular distance formula or graphical verification. Transition to the second example: 'A ugali-delivery lorry is at A(−1, 4) and its image in a new mirror is at A′(3, 0).	Worked Example → Generalisation (Fading Scaffolding): The teacher fully models the first example, partially models the second with cold-calls filling in steps, and learners independently apply the algorithm. This 'I do – We do – You do' structure with explicit generalisation of the algorithm makes the procedure transferable to any object–image pair and builds towards the autonomous congruence-testing work in Lessons 6–7. Connecting each algebraic step to 'what this means for the matatu mirror' maintains the phenomenon thread.	Exit-check for the second example: teacher circulates and checks four specific learner books before the DQB phase. Success criteria: (1) M(1, 2) correctly found; (2) gradient of mirror line = 1 correctly derived; (3) equation y = x + 1 correctly written; (4) at least one verification step shown. Learners who reach y = x + 1 are challenged: 'Can you show that the mirror line y = x + 1 is NOT the same as y = x from Lesson 3? What is different about this lorry's mirror position compared to the standard y = x mirror?'

		Find the mirror line.' Give 4 minutes of independent work time. Cold-call TWO learners for different steps: 'Baraka — what is the midpoint?' [Expected: M(1,2).] 'Amina — what is the gradient of the mirror line?' [Expected: gradient of AA′ = (0−4)/(3−(−1)) = −1; perpendicular gradient = 1.] Consolidate the three-step algorithm on the board and have all learners copy it into their books with a box around it labelled 'Mirror Line Algorithm'.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Use the Percent Equation to Find Part a (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front. In pairs, they review the board's existing sticky notes from Lessons 1–4 and discuss two questions: 'Which of our earlier questions can we now fully answer because we can find the mirror line equation?' and 'What new questions has today's lesson raised?' Each pair writes one 'NOW WE KNOW' card (green) and one 'NEW QUESTION' card (yellow) and places them on the board. The teacher then facilitates a 3-minute whole-class read-around of new additions.	Point to the DQB and say: 'Our driving question asks why the matatu mirror shows a flipped but same-sized image. We have now found a way to locate the mirror line algebraically. Look at our board — which questions from previous lessons can we tick off?' Give pairs 2 minutes to discuss quietly, then say: 'Write your NOW WE KNOW card in one sentence starting with: The mirror line is always...' After pairs post cards, read three aloud, pausing for 10 seconds after each: 'Does everyone agree this is fully answered? Or is there still something we don't know?' For NEW QUESTION cards, prompt: 'Today we found the mirror line from just ONE object–image pair. What if the shape has many vertices? Do we need to use every vertex, or is one pair enough? Write that question if it came up for you.' Record the most mathematically significant new question — likely about minimum information needed — on a fresh card labelled 'L5 New'. Connect explicitly to the driving question: 'Knowing where the mirror is gets us closer to proving the image and	Driving Question Board as Living Knowledge Tracker: The DQB makes the class's growing understanding visible and cumulative. The green/yellow card system distinguishes consolidated knowledge from open questions, preventing learners from treating each lesson as isolated and maintaining the narrative thread of the matatu phenomenon across all eight lessons.	Teacher reads all posted cards to check for misconceptions. Red flags: a 'NOW WE KNOW' card that states 'the mirror line always passes through the origin' (overgeneralisation from Lesson 3 special cases) or a card that confuses 'perpendicular bisector' with 'angle bisector'. If either appears, address it verbally before the Model Building phase begins: 'I see a card that says the mirror line passes through the origin — Lessons 3 and 5 together. Can someone show me a counter-example using today's work?'

		object are congruent — that is our next goal.'		
Model Building Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Use the Percent Equation to Find Part a (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative model — the annotated diagram/summary page they have been building since Lesson 1 — and add a new layer. Using their preferred representation (annotated diagram, concept map, or written algorithm), they incorporate today's key finding: the three-step Mirror Line Algorithm. They must also draw and label a new example entirely of their own invention: they choose any two points A and A′ (not from today's examples), apply the algorithm, find the mirror line, and sketch the result on a mini Cartesian grid. Below their model they write one sentence connecting today's work back to the matatu phenomenon.	Say: 'Open your cumulative model. You have been building this since we first looked at the matatu conductor's mirror. Today add a new section titled: Finding the Mirror Line. Include: (1) the three-step algorithm in your own words; (2) your own invented example with full working; (3) one sentence connecting this to our driving question.' Allow 5 minutes of independent work. Circulate and look for three things: (a) the algorithm is correctly stated (midpoint → perpendicular gradient → equation); (b) the invented example is correctly solved; (c) the phenomenon connection sentence is substantive (not just 'this is about mirrors'). Cold-call ONE learner to share their invented example and phenomenon sentence: 'Odhiambo — read us your example and your matatu sentence.' After Odhiambo shares, ask the class: 'Did anyone invent an example where the mirror line turned out to be vertical or horizontal? What does that tell us about the lorry's position relative to the mirror?' Allow 10 seconds WAIT TIME. Close the lesson by pointing to the DQB: 'Next lesson we will use everything we know about mirror lines and image properties to test whether two triangles are truly congruent — the same question that started our unit: is the lorry's mirror image truly identical to the lorry itself?'	Cumulative Model Revision with Self-Generated Example: Requiring learners to invent their own object–image pair and solve it consolidates procedural fluency while the phenomenon-connection sentence forces conceptual integration. Updating the same model document across all eight lessons creates a visible record of growing understanding and mirrors the NGSS practice of developing and using models — learners are not just practising a procedure but refining an explanatory tool.	Teacher collects or photographs three learner models (one strong, one developing, one struggling) before the next lesson to inform differentiation in Lesson 6. Success criteria for the model addition: (1) three-step algorithm correctly and clearly stated; (2) invented example yields a correct mirror line equation with working shown; (3) phenomenon sentence explicitly mentions 'perpendicular bisector' or 'equidistant' and connects to the matatu mirror or road safety context. Learners who finish early are challenged: 'Can you find the mirror line if I give you a whole triangle ABC and its image A′B′C′? Do you need all three vertex pairs or just one? Try it.'

D. TEACHER REFLECTION

1. During the Predict phase, how many learners correctly identified that the mirror line must pass through the midpoint of PP′? What does this tell me about whether Lesson 4's properties were consolidated — and how should I adjust the opening of Lesson 6 accordingly?

2. In the Observe phase, how many learners confused the gradient of the mirror line with the gradient of PP′ (i.e., failed to take the negative reciprocal)? Is this a sign that the perpendicularity condition needs re-teaching, or was it a one-off arithmetic slip?

3. Did the matatu/lorry context in the Explain phase feel authentic to learners, or did it feel forced? How might I source a more locally vivid scenario (e.g., a Kisumu fish market with mirrors, or a Kericho tea factory quality-control mirror) to make the context stickier for my specific class?

4. Looking at the DQB additions, did learners generate the question about minimum vertex information needed to find the mirror line? If not, what does that tell me about the depth of their conceptual engagement — and how do I seed that question at the start of Lesson 6?

5. Were learners who completed their cumulative models able to write a genuinely substantive phenomenon-connection sentence, or did they default to surface-level statements? What scaffolding (a sentence stem, a worked example sentence) would help the developing learners write more precisely in Lesson 6?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that when point P(2,1) and its image P′(6,5) are plotted on a Cartesian plane, the segment PP′ has a midpoint M(4,3) and a gradient of 1. A line drawn through M with gradient −1 (the negative reciprocal) acts perfectly as the mirror line — every object point and its image are equidistant from this line and the connector is perpendicular to it.
What did I learn?	The mirror line is always the perpendicular bisector of the segment joining an object point to its image. To find its equation: (1) calculate the midpoint M of the object–image segment; (2) find the gradient of the object–image segment and take its negative reciprocal to get the mirror line gradient; (3) substitute M and the gradient into $y - y_1 = m(x - x_1)$ to write the equation. This three-step algorithm works for any object–image pair, including cases where the mirror line is vertical (x = k) or horizontal (y = k).
How does this explain the phenomenon?	The matatu conductor's mirror is positioned so that it is equidistant from every point of the lorry and the corresponding point of the lorry's image — it is the perpendicular bisector of every such pair. This is why the image appears exactly as far behind the mirror surface as the lorry is in front of it. Knowing how to find the mirror line equation algebraically confirms that the mirror's position is not arbitrary: it is the unique line that preserves all distances and angles, which is the geometric reason the image and object are the same size and shape — a foundation for proving congruence in the next lesson.

LESSON 6 (40 minutes): Constructing Reflected Images — Drawing the Mirror Image of a Flag Outline on Plain Paper	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners physically construct reflected images of a Kenya flag outline and irregular polygons on plain paper using ruler and set square, cementing the perpendicular bisector property as a practical construction technique.
Knowledge	– The perpendicular from any object point to the mirror line, extended an equal distance on the other side, locates the unique image point. – Every vertex of an object reflects to a unique, corresponding image vertex across the mirror line. – Joining image points in order produces the reflected image, which is congruent to the object in every measurable dimension.
Skills	– Accurately draw perpendiculars from object vertices to a given mirror line using a ruler and set square, and mark equal distances to locate image points. – Construct the complete reflected image of an irregular polygon by connecting image vertices in order, and verify congruence by comparing lengths and angles.
Attitudes	– Appreciate that precise geometric construction — not estimation — guarantees a trustworthy mirror image, linking mathematical rigour to the reliability of real-world mirrors such as the matatu rear-view mirror. – Develop patience and care in measurement, recognising that every millimetre of inaccuracy produces a distorted image.
Key Inquiry Question	How do we use reflection in day-to-day life?
Purpose in Storyline	Having discovered the four properties of reflection (Lesson 4) and learned to find the mirror line equation (Lesson 5), learners now reverse the process — given an object and a mirror line, they construct the image step by step, proving that the image is always congruent to the object and setting the stage for formal congruence tests in Lessons 7 and 8.
Safety Notes	Rulers and set squares have sharp edges and corners — learners should handle them carefully and keep them flat on the desk. No other specific hazards.

B. LESSON OVERVIEW

This lesson is the practical construction heart of the reflection unit. Learners have spent five lessons building conceptual understanding — lines of symmetry, properties of reflection, coordinate rules, and mirror line equations. Now they apply all of that knowledge with ruler, set square, and pencil on plain paper, constructing reflected images of a Kenya flag outline and at least one irregular polygon entirely from first principles. The anchor phenomenon — the matatu conductor's rear-view mirror on the Nairobi–Thika highway — is revisited explicitly: just as the mirror locates each point of the lorry at an equal perpendicular distance behind the glass, learners locate each image vertex by dropping a perpendicular from the object vertex to the mirror line and marking an equal distance on the other side. The construction process makes the perpendicular bisector property tangible and memorable.

The lesson is structured so that the flag outline (a recognisable, meaningful Kenyan shape) provides a motivating first construction task, and a second irregular polygon (a quadrilateral or pentagon with no special symmetry) challenges learners to generalise the method beyond familiar shapes. Throughout, the teacher emphasises that the image produced is not an approximation — it is geometrically exact — and that measurement of corresponding sides and angles will confirm this congruence. This prepares learners for the formal statement of congruence conditions in the next two lessons.

By the end of the lesson, learners will have constructed at least two reflected images, verified that every corresponding length and angle is preserved, and recorded the key generalisation: "Every point reflects to a unique image point across the mirror line, and the image is always congruent to the object." The Driving Question Board is updated with a learner-drawn labelled diagram showing object, mirror line, and image, making the growing understanding visible to the whole class.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a pre-drawn outline of Kenya's flag shape (a simple rectangular outline with the Maasai shield roughly sketched inside — just enough vertices to make it a polygon, not a detailed flag) and a bold dashed mirror line drawn about 4 cm to its right. Without any tools, each learner sketches in their exercise book where they think the reflected image will appear. They mark with a cross (×) where they believe each of the four or five key vertices will land. Learners then share their prediction with a partner, noting any disagreements about how far from the mirror line each image vertex will be.	Display the prepared worksheet (or draw on the board) showing the flag polygon labelled A, B, C, D, E and the mirror line m. Say: 'Before we pick up any tools, I want your prediction. Where will each vertex land when this flag is reflected in the mirror line? Sketch it — do not measure yet.' Allow 2 minutes of silent individual prediction. After 2 minutes, say: 'Turn to your neighbour and compare. Do you agree on vertex A's image position? What about vertex C?' [WAIT TIME: 15 seconds after each question before taking responses.] Cold-call 3 pairs: 'Pair near the window — where did you put A-image? Why?' 'Pair at the back — do you agree?' 'Pair by the door — what did you disagree on?' Record two or three contrasting predictions as rough sketches on the board under the heading 'Our Predictions.' Say: 'We are going to find out which prediction the mathematics supports — and why.'	Predict–Sketch–Compare: Learners commit to a visual prediction before instruction, activating prior knowledge of the perpendicular bisector property from Lesson 4. Comparing predictions with a partner surfaces misconceptions (e.g., learners who reflect along a diagonal rather than perpendicularly) that the construction activity will directly resolve.	Teacher scans partner sketches for two key misconceptions: (1) learners who place the image vertex at the same horizontal distance rather than the perpendicular distance from the mirror line; (2) learners who 'slide' the image parallel to the mirror line rather than reflecting it. These learners are mentally flagged for targeted questioning during the Observe Phase.
Observe Phase 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners receive a plain-paper worksheet with the flag polygon (vertices A, B, C, D, E clearly labelled, coordinates given as reference points so they can check later) and a vertical mirror line m. Following a teacher-modelled step-by-step demonstration of one vertex, learners work individually to: (1) place a set square against	Before learners begin, model the construction of ONE vertex (vertex A) live on the board or on a large sheet of paper pinned to the wall. Narrate each step aloud: 'I place my set square so one edge lies exactly along the mirror line m. I slide it until the perpendicular edge passes through vertex A. I draw the perpendicular line — this is the	Step-by-Step Construction with Live Modelling: The teacher models one vertex explicitly, then releases learners to replicate the method. This 'I do – You do' structure, combined with the measurement table, ensures learners gather their own numerical evidence that corresponding lengths are	Teacher checks three things while circulating: (1) Is the set square correctly aligned to the mirror line before the perpendicular is drawn? (2) Is the measured distance on the image side exactly equal to the object side distance (within 1 mm)? (3) Are image vertices joined in the correct order, preserving the polygon shape?

📖 **HTML:** [Reflection Symmetry (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the mirror line and draw the perpendicular from vertex A to line m; (2) measure the perpendicular distance from A to m with a ruler; (3) mark the image point A' on the other side at the same measured distance; (4) repeat for all remaining vertices; (5) join A', B', C', D', E' in order to form the complete image polygon. They then use their ruler to measure and record in a table: AB and A'B', BC and B'C', etc., noting whether each pair of corresponding lengths is equal.	construction line, so I draw it lightly. Now I measure from A to where my perpendicular crosses m — I get 3.2 cm. I mark 3.2 cm on the other side and put a dot — that is A-prime.' Then say: 'Now you do vertex B the same way. I am coming around — I want to see your set square correctly aligned before you draw anything.' Circulate for 5 minutes, crouching to eye level with learners. For any learner who has their set square misaligned, say: 'Your set square needs to sit flush along the mirror line — show me the edge touching m first, then rotate until the perpendicular edge reaches your vertex.' [WAIT TIME: 10 seconds after corrective prompt before checking again.] After the first three vertices, pause the class: 'Stop. Hold up your page. Can you see your image starting to take shape?' Cold-call 2 learners: 'Wanjiku — describe what your image looks like so far compared to the object.' 'Otieno — is the image on the correct side of the mirror line?' After all five vertices are located, say: 'Now measure corresponding sides and fill in the table. I want numbers, not estimates.'	preserved — they are not told the result; they discover it through measurement. Connecting each perpendicular to the matatu mirror ('just as the mirror is equally far from lorry and image, your construction line is equally far from vertex and image vertex') keeps the phenomenon alive.	Teacher notes any learner whose image polygon is a different shape or size and provides immediate corrective feedback — most errors trace to a misaligned set square at step 1.
Explain Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Reflection Symmetry (Flexbook)]	Learners complete a second construction task: an irregular quadrilateral PQRS (with no right angles and no equal sides — shaped loosely like a plot of land near Kericho) with a mirror line at an oblique angle (not vertical) already drawn on the worksheet. They apply the same perpendicular bisector construction method independently, then answer three written explanation prompts in their books: (1) 'Explain	Introduce the Kericho land plot polygon with: 'Farmers near Kericho Tea Estates often work with irregular plots of land — no neat right angles, no equal sides. Imagine PQRS is such a plot. The mirror line m is an irrigation channel running diagonally across the field. If we reflect the plot across the channel, where does each corner land?' Give 8 minutes for independent construction and the three written prompts. Then	Explanation Prompts with Phenomenon Re-anchoring: The three written prompts escalate from procedural explanation (why PQ = P'Q') to conceptual transfer (would moving the mirror change size?) to phenomenon application (the matatu lorry). This sequence ensures learners articulate the mathematical justification in their own words before connecting it back to the driving question — building the language of	Teacher reads prompt 3 responses over learners' shoulders during circulation. Target response: 'The image of the lorry is also 8 m long because reflection preserves length — every point of the lorry reflects to a unique image point at equal perpendicular distance from the mirror, and the distance between any two image points equals the distance between the corresponding object points.' Learners who write 'it

Source: CK-12 🔍 Search ARES for similar readings	in your own words why PQ = P'Q' in your construction.' (2) 'If the mirror line were moved 2 cm to the right, would the image change shape or size? Explain.' (3) 'A matatu mirror shows the image of a lorry that is 8 m long. What is the length of the image of the lorry in the mirror? Justify your answer using what you have just constructed.'	facilitate a whole-class debrief. Say: 'Let us hear your answer to prompt 3 — the matatu lorry.' [WAIT TIME: 12 seconds.] Cold-call 3 learners with different ability levels: 'Kamau — what did you write?' 'Achieng — do you agree with Kamau? What would you add?' 'Mwangi — can you connect your construction to the lorry in our matatu scenario?' After responses, say: 'So the image of an 8 m lorry in the mirror is also 8 m — the mirror does not shrink or stretch, it only flips. That is what your construction just proved — every corresponding length is preserved.' Write on the board: 'Reflection PRESERVES length, angle, and shape → Object ≅ Image.' Ask: 'Does anyone remember what the word for this is — when two shapes are identical in every measurable way?' [WAIT TIME: 10 seconds.] Elicit 'congruent.'	congruence they will need in Lessons 7 and 8.	depends on how far the lorry is' have confused reflection with perspective/enlargement and need a one-to-one follow-up question: 'Did the length of your polygon PQRS change when you reflected it? Check your measurement table.'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Proofs with transformations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner selects their neatest construction — either the flag outline or the Kericho land plot — and in the space below it writes a two-sentence caption: one sentence stating what they did, one sentence stating what it proves. Volunteers share their captions aloud. The class agrees on a consensus diagram to add to the class Driving Question Board: a labelled sketch showing object polygon, mirror line m, perpendicular construction lines from two vertices, and the image polygon, with the caption 'Every point reflects to a unique image point — object and image are congruent.' Learners also scan the existing DQB cards from Lessons 1–5 and write on a sticky note one NEW question that today's lesson	Say: 'Look at your best construction. Write underneath it: one sentence — what you did today; one sentence — what it proves about reflection.' Allow 3 minutes. Cold-call 4 learners to read their captions. After each, ask the class: 'Thumbs up if your caption says something similar. Thumbs sideways if you said it differently.' Use the strongest caption (or a composite) to model the consensus DQB entry. Pin the chosen diagram to the DQB and say: 'This diagram answers part of our driving question — yes, the mirror image is always the same shape and size as the object. But look at our driving question again [point to it]: it also asks WHY the image appears flipped and how this proves congruence. We have	Caption Writing and DQB Progressive Update: Requiring learners to write a two-sentence caption (what I did / what it proves) forces consolidation of procedural and conceptual learning simultaneously. The DQB update makes the class's growing understanding visible and cumulative — each lesson's addition builds a public record of the evidence-gathering sequence. The 'new questions' sticky notes maintain intellectual curiosity and preview the congruence test focus of Lessons 7 and 8.	Teacher reads the new-question sticky notes. Expected emerging questions include: 'Does it matter which three sides we compare to call two triangles congruent?' 'Can two shapes be the same size and shape but one is flipped — is that still congruent?' 'What is the minimum information needed to prove congruence?' These questions confirm learners are ready for the congruence test lessons. Learners who write very procedural questions ('How do you use the set square?') may need additional support consolidating the conceptual significance of today's construction.

	raised for them about congruence.	shown congruence through construction — but we have not yet stated the MINIMUM conditions needed to guarantee two triangles are congruent. That is where we are heading.' Then say: 'Write one new question your construction raised — stick it on the DQB under the heading Things We Are Still Wondering.' [WAIT TIME: 2 minutes for silent writing.] Read 3–4 sticky notes aloud, briefly validating each.		
Model Building Phase 📹 **VIDEO:** [Proofs with transformations] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Reflection Symmetry (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative model — the annotated diagram they have been building since Lesson 1, which currently shows: a matatu mirror, a lorry object and its image, the mirror line, and labels for the four properties of reflection. Today they add three new elements: (1) labelled perpendicular construction lines from at least two object points to the mirror line with the equal-distance marks shown; (2) an annotation reading 'Perpendicular bisector property → image point is unique and equidistant'; (3) a new label on the image that reads 'Image ≅ Object (same shape, same size, all lengths and angles preserved).' Learners then write one sentence at the bottom of their model: 'The next thing we need to explain is…' previewing the congruence tests.	Say: 'Open your cumulative model diagram. Today we add the construction machinery that makes the mirror work. I want to see three things added before you pack up — construction lines with equal-distance marks, the perpendicular bisector label, and the congruence symbol with the word congruent spelled out.' Circulate for 4 minutes. For learners who add the construction lines without the equal-distance tick marks, say: 'Your perpendicular is correct — but what do the tick marks tell us? Add them — one tick on each side of the mirror line — to show the distances are equal.' For learners who write only 'same size' in their congruence annotation, prompt: 'What else is preserved besides size? Check your measurement table from the construction.' After 4 minutes, close with: 'Hold up your model. The model is growing — you have moved from seeing the phenomenon [point to matatu/lorry], to explaining the properties, to constructing the image, and now labelling the proof of congruence. Two more lessons and we will have the complete answer to our driving question.' Assign take-home task: 'At home,	Cumulative Model Revision with Annotation Requirements: The three-element addition requirement (construction lines, perpendicular bisector label, congruence symbol) ensures the model grows in specificity and mathematical language with each lesson. The model is not re-drawn from scratch — it is annotated and extended, mirroring how scientific understanding builds incrementally. The 'next thing we need to explain' sentence creates a forward-looking knowledge gap that motivates the congruence test lessons.	Teacher does a rapid visual scan of models held up: check that (1) perpendicular construction lines are drawn (not diagonal or freehand curves), (2) equal-distance tick marks appear on both sides of the mirror line, and (3) the congruence symbol (≅) appears with correct labelling. Models missing any of these three elements are flagged for a brief check-in at the start of Lesson 7. The take-home sketch task provides informal evidence of whether learners can identify lines of symmetry independently in their everyday Kenyan environment.

		find ONE object in your house that has a line of symmetry — a sufuria lid, a window frame, a mandazi — sketch it and mark the line of symmetry. Bring the sketch tomorrow.'		

D. TEACHER REFLECTION
1. During the Observe Phase construction, which specific step caused the most errors — aligning the set square, measuring the perpendicular distance, or marking the image point — and what adjustment to the modelling sequence would reduce that error next time? 2. Did learners successfully connect the perpendicular bisector construction to the matatu mirror phenomenon, or did the construction feel like an isolated geometric technique? What questioning move could strengthen that connection? 3. When learners wrote their Explanation Prompt 3 response (the 8 m lorry), did most learners correctly reason from preserved length, or did several confuse reflection with enlargement/perspective? How should this misconception be addressed before the congruence test lessons? 4. What patterns appeared in the new-question sticky notes added to the DQB — are learners generating questions about minimum congruence conditions, or are they still focused on procedural construction steps? How does this inform the entry point for Lesson 7? 5. Were learners able to independently construct the oblique mirror line reflection of the irregular quadrilateral (the Kericho land plot), or did the non-vertical mirror line cause significant difficulty? If so, should additional scaffolding around oblique perpendiculars be built into the opening of Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that when they dropped perpendiculars from each vertex of the Kenya flag polygon to the mirror line and marked equal distances on the other side, the resulting image vertices — when joined — formed a polygon that was the same shape and size as the original, with every corresponding side length matching exactly in their measurement tables.
What did I learn?	Every object point reflects to a unique image point located by the perpendicular bisector property (equal perpendicular distance on the opposite side of the mirror line); joining image points in order produces an image that is congruent to the object — all lengths and angles are preserved regardless of the shape's complexity or the mirror line's orientation.
How does this explain the phenomenon?	This explains why the matatu conductor's rear-view mirror shows the approaching lorry at exactly the same size as the real lorry — the mirror applies the perpendicular bisector property to every point on the lorry simultaneously, preserving all distances and angles. The image is not a shrunken or distorted copy; it is a congruent copy, which is why every measurable dimension of the lorry in the mirror matches the real lorry perfectly.

LESSON 7 (40 minutes): Congruence Tests for Triangles — Proving the Matatu Mirror Image is Always an Exact Copy	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners establish the four congruence tests for triangles (SSS, SAS, AAS/ASA, RHS) and use them to formally prove that a reflected triangle is always congruent to its object, resolving the core puzzle of the matatu mirror phenomenon.
Knowledge	– State and apply the four congruence tests: SSS (three sides), SAS (two sides and included angle), AAS/ASA (two angles and a side), and RHS (right angle, hypotenuse, and one side). – Explain why a reflected image of a triangle satisfies at least one congruence test with respect to its object. – Distinguish between direct congruence (same orientation) and opposite/indirect congruence (mirror orientation) as seen in the matatu rear-view mirror.
Skills	– Use ruler and pair of compasses (geometrical set) to construct a triangle congruent to a given triangle using each of the four tests. – Verify congruence by measuring corresponding sides and angles and matching them systematically.
Attitudes	– Appreciate that mathematical proof — not just visual appearance — is what guarantees the mirror image of the lorry in the matatu mirror is truly identical in every measurable dimension. – Show responsibility for precise measurement and honest recording of results when carrying out congruence investigations.
Key Inquiry Question	Where do we use congruence in real life, and what is the minimum information needed to guarantee two triangles are identical in shape and size?
Purpose in Storyline	Lessons 1–6 established how reflections work and how to construct and find mirror lines. This lesson completes the evidence-gathering sequence by providing the formal mathematical proof that the mirror image is always congruent to the object — answering the driving question's challenge: why can we guarantee the lorry and its mirror image are always the same shape and size, even though the image is flipped?
Safety Notes	Compasses have sharp points — learners must keep compass points directed downward onto paper at all times and pass geometrical sets with the point-end down. No other specific hazards.

B. LESSON OVERVIEW
This lesson is the seventh and penultimate lesson in the evidence-gathering sequence for Sub-Strand 2.2. In lessons 1–6, learners anchored their investigation in the matatu rear-view mirror phenomenon, discovered properties of reflection, plotted reflected images on the Cartesian plane, determined mirror line equations, and constructed reflected images on plain paper. What has remained unresolved — and explicitly tracked on the Driving Question Board — is the formal proof of why the mirror image is always an exact copy (same shape, same size) of the object. This lesson provides that proof through the four triangle congruence tests: SSS, SAS, AAS/ASA, and RHS.\n\nLearners begin by predicting what minimum information about two triangles would be needed to be certain they are identical, connecting this to the matatu mirror context. They then carry out a hands-on investigation using paper cutouts of triangles, ruler, and compasses, attempting to construct a second triangle that matches a given one using only partial information (three sides; two sides and included angle; two angles and a side; right angle, hypotenuse and side). By comparing their constructions they observe that each test produces a unique triangle — sometimes a direct copy, sometimes a mirror-flipped copy — establishing the four congruence tests empirically before formalising them. In the explain phase learners apply these tests to a reflected triangle on the Cartesian plane to formally prove that a reflected image is always congruent (opposite/indirect congruence) to its object, directly answering the driving question.\n\nThe lesson is intentionally grounded in Kenya-relevant contexts throughout: the matatu mirror phenomenon drives every phase; triangular cutouts are labelled with Kenyan place-name vertices (e.g. triangle NMK for Nairobi–Mombasa–Kisumu); and learners discuss how congruence appears in everyday Kenyan life — identical triangular samosa pastry shapes cut from the

same template, identical triangular roof trusses in Rift Valley buildings, and the tiling patterns on Nairobi pavements. By the end of the lesson learners can state all four congruence tests, apply them to verify that two triangles are congruent, and articulate why the matatu mirror image satisfies SSS — providing the formal mathematical resolution to the unit's anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two triangles drawn on the board — triangle NMK (vertices labelled Nairobi, Mombasa, Kisumu) and its reflected image triangle N'M'K' — and are asked individually to write in their exercise books: 'What is the MINIMUM information about these two triangles that would convince you, without measuring everything, that they are exactly identical?' They record their prediction (a list of measurements or conditions) and share briefly with a shoulder partner before the class shares out.	Display the two triangles (object and mirror image of a scalene triangle with sides 5 cm, 7 cm, 9 cm) on the board without any labels other than vertex names. Say: 'Look carefully at triangle NMK and its mirror image triangle N'M'K' in the matatu's rear-view mirror. In your book, write down the smallest list of measurements you would need to check to be 100% sure these two triangles are exactly identical — same shape, same size. You have 2 minutes working alone.' [WAIT TIME: 2 minutes, silent individual work.] Then say: 'Now share your list with the person next to you. Do you agree? You have 1 minute.' [WAIT TIME: 1 minute.] Cold-call 4 learners to share their predictions. Write each suggestion on the board in a column headed 'Our Predictions.' Accept all responses without evaluation — expected responses include 'measure all three sides,' 'measure all three angles,' 'measure two sides and the angle between them,' 'measure one side only.' Say: 'By the end of today's lesson we will know exactly which of these predictions are correct — and we will use that answer to prove, once and for all, that the lorry in the matatu mirror is always an exact copy of the real lorry.'	Prediction-before-instruction anchored to phenomenon: learners activate prior knowledge about what makes two shapes identical, creating cognitive need for the formal congruence tests. Writing predictions before sharing prevents anchoring bias and ensures every learner commits to a position.	Teacher scans predictions as learners write — look for: (a) learners who list all six elements (3 sides + 3 angles), suggesting they do not yet know the minimum; (b) learners who list 'three sides' or 'two sides and angle' — these may have prior intuition; (c) learners who write 'one side' or 'one angle' only — flag these for targeted questioning during observe phase. Record 3–4 representative predictions on the board for revisiting at the end of the lesson.

Observe Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Working in pairs, learners carry out four short construction tasks using ruler and compasses. Each pair receives a printed or hand-drawn reference triangle (sides 5 cm, 7 cm, 9 cm; scalene) and attempts to construct a new triangle using only the information given for each test: Task A — given all three side lengths (SSS); Task B — given two sides (5 cm and 7 cm) and the included angle between them (SAS); Task C — given two angles (at Nairobi vertex 40° and Mombasa vertex 60°) and the side between them (ASA); Task D — given it is right-angled, hypotenuse 9 cm, and one leg 5 cm (RHS). After each construction, pairs measure the remaining sides/angles of their new triangle and compare it to the reference. They record: 'Does my constructed triangle match the reference exactly? How many different triangles could I have made with this information?'	Distribute geometrical sets (ruler, compass, protractor). Say: 'You are going to try to build a copy of triangle NMK using only partial information — just like the matatu driver can only see part of the lorry in the mirror. I will give you four tasks. For each one, try to construct the triangle, then measure what you did not know and check if it matches our original.' [WAIT TIME: Allow 12 minutes for all four construction tasks.] Circulate actively — at each pair, ask: 'How many different triangles do you think you could draw with just this information?' [WAIT TIME: 10 seconds after asking, before accepting answer.] After 12 minutes, bring class together. Cold-call 3 pairs per task to report findings. Write on board in a table with columns: Task | Information used | Does it give a unique triangle? | Name of test. Guide learners to fill in the table collectively, highlighting that each of the four tasks produces exactly one triangle (up to flipping). If time permits, demonstrate on the board the SSS construction step-by-step. Say: 'Notice that in Task A and Tasks B, C, D, your triangle is an exact copy of mine — but sometimes it is flipped. That flipped copy is exactly what the matatu mirror produces. We call this opposite congruence.'	Hands-on construction investigation (NGSS SEP-3: Planning and Carrying Out Investigations): learners generate their own empirical evidence that each test produces a unique triangle. Recording results in a structured table builds the habit of organising evidence before drawing conclusions, directly mirroring how engineers verify that manufactured parts (e.g. identical roof trusses) are congruent.	Teacher checks constructions during circulation: (a) Does the pair's SSS construction, when measured, match all three sides of the reference? (b) Does the SAS construction match? (c) Are learners correctly using the compass to mark off side lengths (not estimating)? Flag pairs who measure angles with a ruler — redirect: 'Use your protractor for angles, your compass and ruler for sides.' After class reporting, check that all four test names appear correctly in the board table — if any are missing or mislabelled, prompt the class with: 'What do the letters S, A, R, H stand for?'
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES	Learners individually apply the four congruence tests to a worked example: triangle PQR with P(1,1), Q(5,1), R(5,4) and its reflection triangle P'Q'R' in the y-axis with P'(−1,1), Q'(−5,1), R'(−5,4). They calculate the lengths of all three sides of both triangles using the	Write triangle PQR and P'Q'R' on the board (or display on screen) with coordinates. Say: 'Now we will use what we just discovered to formally prove — using mathematics, not just our eyes — that the matatu mirror image of any triangle is always congruent to	Worked example with formal congruence statement (NGSS SEP-6: Constructing Explanations): moving from empirical observation (construction tasks) to formal mathematical proof (congruence statement with symbol and test name) deepens	Collect one exercise book per pair and quickly check: (a) Are side lengths calculated correctly (PQ = 4, QR = 3, PR = 5; P'Q' = 4, Q'R' = 3, P'R' = 5)? (b) Is the congruence test identified as SSS? (c) Is the congruence statement written with the ≅ symbol and correct vertex

for similar videos 📖 **HTML:** Properties of Equality and Congruence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	distance formula (or Pythagoras on the grid), list corresponding sides, and write a formal congruence statement: 'Triangle PQR ≅ Triangle P'Q'R' by SSS.' They also identify whether this is direct or opposite congruence and explain in one sentence why it is opposite. Finally, in a class discussion, they connect this back to the matatu: 'The lorry and its mirror image satisfy SSS congruence, which is why the image is always exactly the same size and shape — but because the congruence is opposite, the right headlight appears on the left.'	the original. Work through this example in your book. First, calculate PQ, QR, PR. Then calculate P'Q', Q'R', P'R'. Then decide which congruence test applies.' [WAIT TIME: 5 minutes individual work.] Cold-call 2 learners to state the side lengths — write on board. Ask: 'Which congruence test applies?' [WAIT TIME: 10 seconds.] Cold-call 1 learner. Say: 'Write the congruence statement in your book using the symbol ≅.' After learners write it, ask: 'Is this direct or opposite congruence — can you pick up triangle PQR and place it exactly on top of P'Q'R' without flipping it off the page?' [WAIT TIME: 10 seconds.] Cold-call 1 learner. Summarise: 'Every reflected triangle is congruent to its object by SSS — and it is always opposite (indirect) congruence, because reflection flips the orientation. This is the mathematical proof that the lorry in the matatu mirror is an exact copy — same sides, same angles — but mirror-reversed.' Write the four tests in a summary box on the board: SSS, SAS, AAS/ASA, RHS.	understanding. Connecting the abstract proof back to the matatu phenomenon in the final sentence ensures every learner can articulate the real-world meaning of the mathematics.	order? (d) Is the direct/opposite distinction correctly identified as 'opposite'? Note any learner who writes SAS or AAS — probe: 'You found all three sides equal — which test uses three sides?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of	In small groups of three, learners review the class Driving Question Board. They are given two sticky notes each. On the first (green), they write one claim their group can now confidently make to answer part of the driving question: e.g. 'We can prove the mirror image of the lorry is congruent to the lorry by SSS.' On the second (orange), they write one question that Lesson 7 has raised or left unanswered for them: e.g. 'Does SSS always work for non-triangles — like the	Direct learners' attention to the DQB. Say: 'Look at the questions we have posted since Lesson 1. Which ones can we now answer — or partially answer — after today?' [WAIT TIME: 30 seconds for learners to scan the board.] Distribute sticky notes. Say: 'Green note: write one thing you can now PROVE about the matatu mirror image, using the congruence tests from today. Orange note: write one question this lesson has raised for you that we have not yet fully answered.' [WAIT TIME: 3	Claim–Evidence–Reasoning on the DQB (NGSS SEP-7: Engaging in Argument from Evidence): separating proven claims (green) from new questions (orange) trains learners to distinguish what is now established from what remains open, building the habit of evidence-based reasoning that underpins both mathematical proof and scientific inquiry.	Read all green sticky notes — look for: (a) Claims that correctly name a congruence test (SSS, SAS, AAS, RHS); (b) Claims that use the word 'prove' or 'congruent'; (c) Claims that reference the matatu/mirror context. Flag any green note that makes an incorrect claim (e.g. 'two sides are enough') — use it as a teaching moment without embarrassing the group: 'This is interesting — can the class find the counterexample to this claim?'

Equality and Congruence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	rectangular lorry body?' Groups place their sticky notes on the DQB and the class does a brief gallery read of the new additions.	minutes.] Ask pairs to come up and stick notes. Cold-call 3 groups to read their green claim aloud. After each, ask the class: 'Do you agree? Give a thumbs-up if you accept this as a proven claim.' For any contested claim, ask: 'What evidence from today supports or challenges this?' Draw attention to any orange notes that connect to Lesson 8 (the final lesson), e.g. questions about applying congruence to real-life engineering — say: 'These are great questions. Our final lesson will explore exactly how congruence is used in design and construction — and how it connects back to our matatu mirror.'		
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Equality and Congruence (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative model — the annotated Cartesian plane diagram they have been building since Lesson 3, showing the matatu mirror phenomenon with object and image triangles. They now add a formal congruence proof box alongside their diagram: they list the three pairs of equal sides (with measurements), state the congruence test used (SSS), write the congruence statement (e.g. △NMK ≅ △N'M'K'), and annotate the diagram with tick marks on corresponding equal sides. They also add a label: 'Opposite (indirect) congruence — orientation is flipped, like the lorry's headlights in the mirror.' Learners who finish early attempt an extension: identify which congruence test would apply if only two sides and the included angle of the object and image were given (SAS), and explain why this also confirms congruence.	Say: 'Open your cumulative model from Lesson 3. Today you are going to add the most important layer yet — the formal mathematical proof that seals our answer to the driving question.' Display a model example on the board showing where the proof box should go (adjacent to the diagram, connected with an arrow). Say: 'Add tick marks on your diagram to show equal sides — one tick on the shortest sides, two ticks on the medium sides, three ticks on the longest sides — just like engineers mark identical parts in a blueprint.' [WAIT TIME: 5 minutes for model revision.] Circulate and ask targeted questions: 'Which sides are corresponding? How do you know?' [WAIT TIME: 10 seconds per learner before moving on.] Cold-call 2 learners to hold up their models. Ask the class: 'Does this model now answer our driving question? What piece — if any — is still missing?' Foreshadow	Cumulative model revision with formal annotation (NGSS SEP-2: Developing and Using Models): adding a proof box with tick marks and a congruence statement transforms the model from a visual record of observations into a mathematical argument. This makes the model an explanatory tool, not just a descriptive one — learners can now use it to predict what will happen with any reflected triangle.	Teacher checks models during circulation for four elements: (1) tick marks correctly placed on corresponding equal sides; (2) congruence statement written with ≅ symbol and correct vertex correspondence; (3) test name (SSS) stated; (4) 'opposite congruence / orientation flipped' annotation present. Any model missing element (4) — the direct/opposite distinction — gets a prompt: 'Can you pick up and place your object triangle directly onto the image without lifting it off the page? What does that tell you about the type of congruence?'

		Lesson 8: 'In our final lesson we will look at how architects and engineers in Kenya use congruence — and how the matatu mirror connects to road safety. Make sure your model is complete and labelled clearly.'		

D. TEACHER REFLECTION
1. During the construction tasks (Observe Phase), did all learners correctly use the compass to mark off side lengths rather than estimating by eye? If not, what additional scaffolding on compass technique is needed before Lesson 8?\n2. When learners wrote their formal congruence statements in the Explain Phase, did the majority use the correct vertex correspondence (e.g. △PQR ≅ △P'Q'R' rather than △PQR ≅ △Q'P'R')? If vertex order errors were common, how will I address this explicitly in the next lesson?\n3. Did learners successfully distinguish between direct and opposite (indirect) congruence? Were they able to connect 'opposite congruence' back to the observed phenomenon of the lorry's headlights appearing reversed in the matatu mirror?\n4. Looking at the green sticky notes on the DQB, how many groups produced a claim that both names the congruence test AND links it to the matatu mirror phenomenon? What does the proportion of such claims tell me about the depth of conceptual integration achieved?\n5. Were the extension learners (SAS task) able to articulate why SAS also guarantees congruence — or did they simply accept it as a rule? How can I use their responses to deepen the whole class's understanding of why the minimum conditions are sufficient?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Through compass-and-ruler construction tasks, learners observed that four specific sets of minimum information — three sides (SSS), two sides and an included angle (SAS), two angles and a side (AAS/ASA), and a right angle with hypotenuse and one side (RHS) — each produce exactly one unique triangle (up to orientation). They also observed that a reflected triangle always has the same three side lengths as its object, and that the reflection produces a mirror-flipped (opposite) copy.
What did I learn?	Two triangles are congruent if they satisfy any one of four tests: SSS, SAS, AAS/ASA, or RHS. A reflected triangle is always congruent to its object by SSS (all three pairs of corresponding sides are equal, since reflection preserves all distances). This is always opposite (indirect) congruence because reflection reverses orientation. The four tests give the minimum conditions needed to guarantee congruence — it is not necessary to measure all six elements (three sides and three angles).
How does this explain the phenomenon?	The matatu rear-view mirror phenomenon is now formally resolved: the image of the lorry in the mirror is congruent to the real lorry (by SSS) because reflection preserves all distances — every side of the lorry maps to an equal side in the image. The image is the same size and shape in every measurable dimension. The apparent 'reversal' (right headlight appearing on the left) is not a size or shape change — it is a flip of orientation, which is the defining property of opposite (indirect) congruence. This is why the lorry and its mirror image are always identical in every measurement, yet cannot be superimposed without flipping.

LESSON 8 (40 minutes): Final Explanation — Unifying Reflection and Congruence to Solve the Matatu Mirror Mystery	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all eight lessons of the sub-strand to construct a complete, written and diagrammatic explanation of why the matatu rear-view mirror produces an image that is the same size and shape as the real lorry yet appears laterally flipped, using coordinate reflection rules and congruence tests as formal proof.
Knowledge	– Reflection maps every object point to a unique image point according to a specific algebraic coordinate rule determined by the mirror line (y-axis: (a,b)→(−a,b); x-axis: (a,b)→(a,−b); y=x: (a,b)→(b,a); y=−x: (a,b)→(−b,−a); y=k: (a,b)→(a,2k−b); x=k: (a,b)→(2k−a,b)). – The mirror line is always the perpendicular bisector of the segment joining any object point to its image, and its equation can be determined from coordinates of object–image pairs. – Two triangles are congruent if and only if they satisfy one of the four tests — SSS, SAS, AAS/ASA, or RHS — and a reflected triangle always satisfies SSS with its object, confirming opposite (indirect) congruence.
Skills	– Construct a complete written explanation linking coordinate transformation rules, mirror-line equations, and congruence tests to the matatu mirror phenomenon. – Compare an initial Lesson 1 model of the phenomenon with a final revised model, annotating every improvement with precise mathematical reasoning.
Attitudes	– Appreciate that mathematics provides exact, universal tools — coordinate rules and congruence tests — that explain everyday phenomena such as driving mirrors, and that this understanding supports road safety. – Value the iterative process of building, testing, and revising a scientific/mathematical model over multiple lessons.
Key Inquiry Question	How do we use reflection and congruence together to fully explain why the matatu mirror image is flipped yet always an exact copy of the original?
Purpose in Storyline	This final lesson closes the inquiry arc: learners move from initial puzzlement about the flipped-but-identical mirror image to a complete, formal explanation grounded in coordinate reflection rules (Lessons 3–6) and congruence tests (Lesson 7), and they revise their Lesson 1 naive model into a mathematically rigorous final model that fully answers the driving question.
Safety Notes	No specific hazards. If small plane mirrors are used during the lesson recap, remind learners to handle glass mirrors carefully and not to direct reflected sunlight into peers' eyes.

B. LESSON OVERVIEW

This is the capstone lesson of the Reflection and Congruence sub-strand. Over the previous seven lessons, learners have investigated symmetry, plotted reflections on the Cartesian plane, discovered the four properties of reflection, determined mirror-line equations, constructed reflected images on plain paper, and applied congruence tests to triangles. In this lesson all those threads are woven into a single, coherent Final Explanation that directly answers the driving question: why does a matatu's rear-view mirror show an image that is the same size and shape as the real lorry yet appears flipped — and how can we use this to prove that the original and its mirror image are always congruent?

Learners begin by revisiting their very first (Lesson 1) prediction sketches and written ideas about the mirror phenomenon, then work in groups to construct a formal written and diagrammatic final explanation that incorporates coordinate rules, the perpendicular-bisector property of the mirror line, and a chosen congruence test. The

Driving Question Board is completed for the last time: every sticky-note question posted in earlier lessons is reviewed, and learners confirm which have now been answered and celebrate the journey of growing understanding. A whole-class gallery comparison of Lesson 1 models versus final models makes the intellectual growth visible.

The lesson closes with a brief road-safety connection — the matatu conductor's use of mirrors is not just a mathematical curiosity but a life-saving application of reflection and congruence — and with teacher-led consolidation of the six coordinate rules as a reference card learners can keep. The teacher reflection questions guide the teacher to evaluate the quality of learners' final explanations and the coherence of the full storyline across all eight lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner retrieves (or is handed back) their Lesson 1 prediction sketch — the drawing they made of what they thought was happening inside the matatu rear-view mirror. Working individually for 3 minutes, learners annotate their original sketch with two colours: green pen to mark anything they got right in Lesson 1, and red pen to mark anything they now know is incomplete or incorrect. They write one sentence summarising what they were most wrong about at the start of the unit.	Display the anchoring phenomenon image on the board — the matatu conductor on the Nairobi–Thika highway holding the small rectangular mirror so the driver sees the approaching lorry. Say exactly: 'Before we look at anything new today, I want you to travel back in time to Lesson 1. Take your original prediction sketch. Give yourself 3 minutes of silence to re-read it and annotate it in two colours.' Set a visible timer for 3 minutes. After the timer sounds, cold-call 3 learners (one from each ability band) to share one thing they marked in red: 'Akinyi — what did you get wrong in Lesson 1 that you now know better?' WAIT TIME: 12 seconds after each name before accepting the answer. Write the three 'misconceptions we started with' on the board as a list. Tell the class: 'These are our starting point. By the end of this lesson, each of those should be fully resolved.'	Elicit–Confront–Resolve (ECR): surfacing prior conceptions held in Lesson 1 and juxtaposing them with current understanding creates the cognitive contrast that makes final consolidation meaningful. Writing in two colours forces learners to actively distinguish what has changed in their thinking rather than passively reviewing notes.	Teacher scans annotated sketches as learners work — look for: (1) red annotations that identify at least one specific misconception (e.g., 'I thought the image was smaller' or 'I did not know the image is the same distance behind the mirror'); (2) the ability to name something correct from Lesson 1. Flag learners whose sketches show no red annotation — they may be avoiding self-critique and will need prompting during group work.
Observe Phase 📹 **VIDEO:** Properties of Equality and Congruence	Groups of four receive a 'Final Evidence Pack' — a single A3 sheet containing: (a) a Cartesian grid showing a lorry outline (quadrilateral ABCD with given coordinates) reflected in the line x	Distribute the A3 Evidence Pack face-down. Say: 'Turn it over — read everything on this sheet silently for 4 minutes. You are detectives. You are looking for the mathematical proof that explains	Evidence-Based Argumentation: learners use a structured evidence pack rather than recall, mirroring the practice of a Kenyan engineer reviewing technical data. By requiring a written justification of	Teacher listens for groups that correctly identify SSS and articulate 'opposite congruence' (the image is flipped, so it cannot be overlaid without lifting). Red flag: groups that say 'they look the

Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	= 0 (the y-axis, representing the mirror surface), with all object and image coordinates labelled; (b) a table of all six coordinate reflection rules studied across Lessons 3–6; (c) measurements confirming AB = A'B', BC = B'C', CD = C'D', DA = D'A'. Learners read and annotate the sheet in silence for 4 minutes, then discuss in their group for 3 minutes: 'Which of the four congruence tests can we use to prove triangle ABC ≅ triangle A'B'C'? Why SSS and not just visual inspection?' Groups record their chosen test and justification in their exercise books.	the matatu mirror.' Set timer for 4 minutes. After silent reading, say: 'Now discuss in your group. You have 3 minutes. Answer this one question: which congruence test proves the lorry and its mirror image are identical? Write your group's answer in your book.' Circulate and listen; do NOT answer questions — redirect with: 'What does the evidence pack say about the side lengths?' After group discussion, cold-call 2 groups to share: 'Kamau's group — which test and why?' WAIT TIME: 15 seconds. Confirm SSS is sufficient (all three corresponding sides are equal because reflection preserves length) and note that this is opposite/indirect congruence because the image cannot be superimposed on the object without lifting it off the page — just as the lorry cannot drive through the mirror.	the congruence test choice, the strategy forces connection between the observed measurements and the formal mathematical claim — a core NGSS Science and Engineering Practice (Constructing Explanations and Designing Solutions).	same' without citing specific equal side measurements — these groups need a targeted question: 'Can you point to the three pairs of equal sides on your grid?'
Explain Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group constructs a 'Final Explanation Poster' on plain A3 paper. The poster must include four labelled sections: (1) The Phenomenon — a sketch of the matatu mirror setup with the object lorry and its image labelled; (2) The Coordinate Rule — the algebraic rule used (reflection in x = 0: (a,b)→(−a,b)), with one worked numeric example using the lorry's coordinates; (3) The Mirror Line — its equation, and proof that it is the perpendicular bisector of at least one object–image segment (midpoint calculation shown); (4) The Congruence Proof — the chosen test (SSS), three pairs of equal sides named, and one sentence explaining why this is opposite congruence. Learners write a 'headline sentence' at the	Give explicit instructions: 'Your poster has exactly four sections — I have written them on the board. Every section must have a diagram AND words. You have 10 minutes.' Write the four section headings on the board with a timer visible. Circulate and use targeted prompts: For Section 2 — 'What is the coordinate rule for reflection in the y-axis? Can you show me with numbers from your grid?' For Section 3 — 'How do you find the midpoint between A(3,4) and A'(−3,4)? What does that midpoint tell you about the mirror line?' For Section 4 — 'Why can't you flip the lorry's image back onto the lorry without lifting it? What is that called?' At the 8-minute mark, say: 'Two minutes left — write your headline sentence first if you have	Structured Poster Construction with Gallery Walk: the four-section scaffold prevents superficial responses and ensures every mathematical layer (phenomenon → coordinate rule → mirror line → congruence) is explicitly connected. The Gallery Walk adds peer accountability and exposes learners to multiple representations of the same idea — a powerful consolidation strategy aligned with NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information).	Teacher evaluates poster sections against three criteria: (1) Does the coordinate rule include a numeric example? (2) Is the midpoint calculation for the mirror line shown step-by-step? (3) Does the congruence section name three specific equal side pairs? Collect two posters (from the strongest and the weakest group) for written feedback to be returned in the next sub-strand. Listen during Gallery Walk for misconceptions voiced aloud — e.g., 'I thought the image was bigger' — and address immediately.

	top of the poster that directly answers the driving question in plain language.	not done so.' After 10 minutes, conduct a 4-minute Gallery Walk: post all posters around the room; each group silently reads two other groups' posters and places a green dot sticker on the most clearly explained section and a question-mark sticker anywhere they are confused. Cold-call one learner per poster to read the headline sentence aloud — WAIT TIME: 10 seconds.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front since Lesson 1. Each learner receives two sticky notes. On the first (blue), they write one question from the board that they can now fully answer, and they write the answer. On the second (yellow), they write one question from the board that they feel is still partially unanswered or leads them to wonder something new. Learners post their notes on the board under two columns: 'Answered ✓' and 'New Wondering 🌟'. The class reads the board together for 2 minutes.	Point to the DQB and say: 'This board has been with us since the very first lesson. Look at every question we posted. Some were posted by you, some were posted together.' Read three specific questions from the board aloud — choose one from Lesson 1, one from Lesson 4, one from Lesson 7. Then say: 'Take your two sticky notes. Blue note: choose one question you can now fully answer — write the question and write the answer in your own words. Yellow note: write one new wondering — something this unit made you curious about that we did not fully explore.' Give 3 minutes for writing. Cold-call 3 learners to read their blue note answers aloud — WAIT TIME: 12 seconds. After each, ask the class: 'Does the class agree that question is answered? Thumbs up if yes.' For the yellow notes, read 3 aloud and tell the class: 'These new wonderings are the beginning of your next mathematics unit — keep them.' Finally, say: 'Our driving question was: why does the matatu mirror show an image that is flipped yet always congruent? Point to the board — together let us say the answer.' Lead a choral response based on the class's	DQB Closure with Answered/Wondering Sort: explicitly sorting the board into 'Answered' and 'New Wondering' columns validates the learning journey and models scientific practice — real mathematicians and engineers answer one question and generate three more. The choral final answer builds shared ownership of the explanation and reinforces the vocabulary (reflection, coordinate rule, perpendicular bisector, congruence, opposite congruence) in a low-stakes oral format.	Teacher reads every blue sticky note posted — look for: correct, complete answers that use at least two technical terms (e.g., 'The mirror line is the perpendicular bisector of the segment joining any object point to its image'). Flag blue notes that are vague ('it goes to the other side') — these learners need individual follow-up. Count the proportion of yellow 'New Wondering' notes that connect to the next sub-strand (Rotation) or to real-life applications — high proportion indicates learners are thinking beyond the immediate task.

		collective headline sentences from the posters.		
Model Building Phase 📹 **VIDEO:** Properties of Equality and Congruence Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Reflection Symmetry (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their exercise books to their Lesson 1 model — the initial sketch of what they thought happened in the matatu mirror. They draw a box around it labelled 'Model 1 — Lesson 1'. Immediately below, they draw their 'Final Model — Lesson 8'. The final model must include: a labelled diagram of the matatu mirror setup (object lorry, mirror line, image lorry); the coordinate rule for reflection in the mirror line; the perpendicular bisector property annotated on one object–image segment; and the congruence test (SSS) with the three equal side pairs labelled. In the final two lines of their exercise book entry, learners write: 'My model improved because…' and complete the sentence with at least two specific mathematical ideas gained across the unit.	Say: 'The last thing we do today is the most important. You are going to look your Lesson 1 self in the eye and show how far you have come.' Allow 6 minutes for individual model drawing — circulate and use encouraging, specific prompts: 'Wanjiku — I can see your coordinate rule. Can you now add the perpendicular bisector annotation? Remember Lesson 5.' 'Otieno — your diagram is clear. Which three sides did you label for SSS?' At the 5-minute mark, cold-call two learners to share one improvement from their 'My model improved because…' sentence — WAIT TIME: 12 seconds. Write their improvement statements verbatim on the board as a class record. Conclude with: 'The matatu conductor on the Nairobi–Thika highway uses this mathematics every single day — he just does not know he is using SSS and perpendicular bisectors. You now know. That is the power of mathematics.' Assign a closing exit ticket (written on a half-sheet): 'State the coordinate rule for reflection in the line y = x, then write one sentence connecting this rule to the matatu mirror phenomenon.' Collect as learners leave.	Model Comparison (Initial vs. Final): placing Model 1 and the Final Model side-by-side in the same exercise book makes intellectual growth concrete and personal. The structured 'My model improved because…' sentence requires metacognitive reflection — learners must identify specific knowledge gained, not just say 'I know more now.' This aligns with NGSS Practice 2 (Developing and Using Models) and the KICD emphasis on learning to learn as a core competency.	Collect exit tickets as learners leave. Successful exit tickets state the rule (a,b)→(b,a) for y=x AND connect it to the phenomenon (e.g., 'If the matatu mirror were tilted at 45°, the lorry's image would have its coordinates swapped — it would appear above rather than to the left'). Exit tickets that only state the rule without a phenomenon connection indicate the learner can reproduce procedures but has not achieved full conceptual transfer — plan a brief review at the start of the Rotation sub-strand for these learners. Final models in exercise books are reviewed during marking: check for all four required components (diagram, coordinate rule, perpendicular bisector annotation, SSS labelling).

D. TEACHER REFLECTION

1. Did every group's Final Explanation Poster include all four required sections with both a diagram and a written explanation? Which section was most frequently incomplete, and what does that reveal about where the sub-strand's instructional sequence needs strengthening in future?

2. When learners annotated their Lesson 1 sketches in red during the Predict Phase, what were the most common misconceptions? Were these misconceptions explicitly addressed in Lessons 2–7, or do they suggest a gap in the storyline that should be closed by adding or reordering a lesson?

3. During the DQB closure, how many 'New Wondering' yellow notes connected meaningfully to the next mathematical topic (Rotation, Sub-Strand 2.3)? If fewer than

half did, consider making the bridge to rotation more explicit in the closing teacher talk.

4. How well did learners use formal mathematical vocabulary (perpendicular bisector, opposite congruence, coordinate rule) in their headline sentences and verbal responses? If vocabulary use was weak, what low-stakes daily routines (e.g., 'word of the day' or 'turn-and-talk with the key term') could be embedded throughout the next sub-strand?

5. Were there learners who could reproduce coordinate rules on the exit ticket but failed to connect them to the phenomenon? What differentiated strategy — perhaps a structured 'Because…Therefore…This means that…' sentence frame — would help these learners achieve conceptual transfer rather than procedural fluency alone?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed their own Lesson 1 prediction sketches alongside a completed Cartesian evidence pack showing the lorry ABCD reflected in the y-axis (x = 0), with all coordinates, side lengths, and the perpendicular bisector of one object–image segment labelled — confirming that the image is the same distance behind the mirror as the object is in front, that all side lengths are preserved, and that the image is laterally flipped.
What did I learn?	Reflection maps every point according to a specific algebraic coordinate rule determined by the mirror line; the mirror line is always the perpendicular bisector of the segment joining any object point to its image; a reflected triangle is congruent to its object by SSS and the congruence is opposite (indirect) because the image cannot be overlaid on the object without lifting it — which is why the lorry's headlights appear on opposite sides in the mirror.
How does this explain the phenomenon?	The matatu rear-view mirror acts like the y-axis of a coordinate plane: it maps every point on the approaching lorry to a unique image point directly opposite, at equal distance, according to the rule $(a, b) \rightarrow (-a, b)$. This preserves all lengths and angles, making the image congruent to the lorry by SSS. The lateral flip — the lorry's right headlight appearing on the left — is the signature of opposite (indirect) congruence: the image is a perfect mathematical copy, but it is a mirror copy, not a direct overlay, which is exactly what any flat reflective surface must produce.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.